

# 10

## Properties of Gases and Gas Mixtures

### Learning Objectives

*After reading this chapter you should be able to*

- LO 10.1 Define Avogadro's law
- LO 10.2 Discuss the equation of state and derive the concept of universal gas constant
- LO 10.3 Summarize various equations of state of an ideal gas
- LO 10.4 Relate specific heat with internal energy and enthalpy of an ideal gas to derive an expression of Gibb's function of inert ideal gases mixture
- LO 10.5 Derive entropy change of an ideal gas and a relation between  $p$ ,  $v$  and  $T$  in a reversible adiabatic process
- LO 10.6 Derive an expression to calculate work transfer in a reversible isothermal process
- LO 10.7 Define polytropic process and discuss its significance
- LO 10.8 Explain the equations of state proposed for real gases
- LO 10.9 Derive virial equations of state in terms of compressibility factor  $Z$
- LO 10.10 Discuss the law of corresponding states and discuss the making of generalized compressibility chart
- LO 10.11 Explain other equations of state including van der Waals and Beattie-Bridgeman
- LO 10.12 Explain Dalton's law of partial pressure and Amagat's law of partial volume in a gas mixture
- LO 10.13 Define Gibb's theorem of gas mixtures applicable for many thermodynamic properties

**10.1 AVOGADRO'S LAW**

A mole of a substance has a mass numerically equal to the molecular weight of the substance.

One g mol of oxygen has a mass of 32 g, 1 kg mol of oxygen has a mass of 32 kg, 1 kg mol of nitrogen has a mass of 28 kg, and so on.

Avogadro's law states that the volume of a g mol of all gases at the pressure of 760 mm Hg and temperature of 0°C is the same, and is equal to 22.4 litres. Therefore, 1 g mol of a gas has a volume of  $22.4 \times 10^3 \text{ cm}^3$  and 1 kg mol of a gas has a volume of  $22.4 \text{ m}^3$  at normal temperature and pressure (N.T.P.).

For a certain gas, if  $m$  is its mass in kg and  $\mu$  is its molecular weight, then the number of kg moles of the gas,  $n$ , would be given by

$$n = \frac{m \text{ kg}}{\mu \frac{\text{kg}}{\text{kg mol}}} = \frac{m}{\mu} \text{ kg moles}$$

The molar volume,  $\bar{v}$ , is given by

$$\bar{v} = \frac{V}{n} \text{ m}^3/\text{kg mol}$$

where  $V$  is the total volume of the gas in  $\text{m}^3$ .

LO 10.1

Define Avogadro's law

**10.2 EQUATION OF STATE OF A GAS**

The functional relationship among the properties, pressure  $p$ , molar or specific volume  $v$  and temperature  $T$ , is known as an equation of state, which may be expressed in the form,

$$f(p, v, T) = 0$$

If two of these properties of a gas are known, the third can be evaluated from the equation of state.

It was discussed in Chapter 2 that gas is the best-behaved thermometric substance because of the fact that the ratio of pressure  $p$  of a gas at any temperature to pressure  $p_t$  of the same gas at the triple point, as both  $p$  and  $p_t$  approach zero, approaches a value independent of the nature of the gas. The ideal gas temperature  $T$  of the system at whose temperature the gas exerts pressure  $p$  (Section 2.5) is defined as

$$T = 273.16 \lim_{p_t \rightarrow 0} \frac{p}{p_t} \quad (\text{Const. } V) \quad (10.1)$$

$$T = 273.16 \lim_{p \rightarrow 0} \frac{V}{V_t} \quad (\text{Const. } p)$$

The relation between  $p$  and  $v$  of a gas may be expressed by means of a power series of the form

$$pv = A(1 + B'p + C'p^2 + \dots) \quad (10.1)$$

LO 10.2

Discuss the equation of state and derive the concept of universal gas constant



where  $A$ ,  $B'$ ,  $C'$ , etc. depend on the temperature and nature of the gas.

A fundamental property of gases is that  $\lim_{p \rightarrow 0} (pv)$  is independent of the nature of the gas and depends only on  $T$ . This is shown in Fig. 10.1, where the product  $pv$  is plotted against  $p$  for four different gases in the bulb (nitrogen, air, hydrogen and oxygen) at the boiling point of sulphur, at steam point and at the triple point of water. In each case, it is seen that as  $p \rightarrow 0$ ,  $pv$  approaches the same value for all gases at the same temperature. From Eq. (10.1)

$$\lim_{p \rightarrow 0} pv = A$$

Therefore, the constant  $A$  is a function of temperature only and independent of the nature of the gas.

$$\lim_{p \rightarrow 0} \frac{p}{p_t} (\text{Const. } V) = \lim_{p \rightarrow 0} \frac{pV}{p_t V} = \frac{\lim_{p \rightarrow 0} pv}{\lim_{p \rightarrow 0} (pv)_t} = \frac{A}{A_t}$$

$$\lim_{p \rightarrow 0} \frac{V}{V_t} (\text{Const. } p) = \lim_{p \rightarrow 0} \frac{pV}{p V_t} = \frac{\lim_{p \rightarrow 0} pv}{\lim_{p \rightarrow 0} (pv)_t} = \frac{A}{A_t}$$

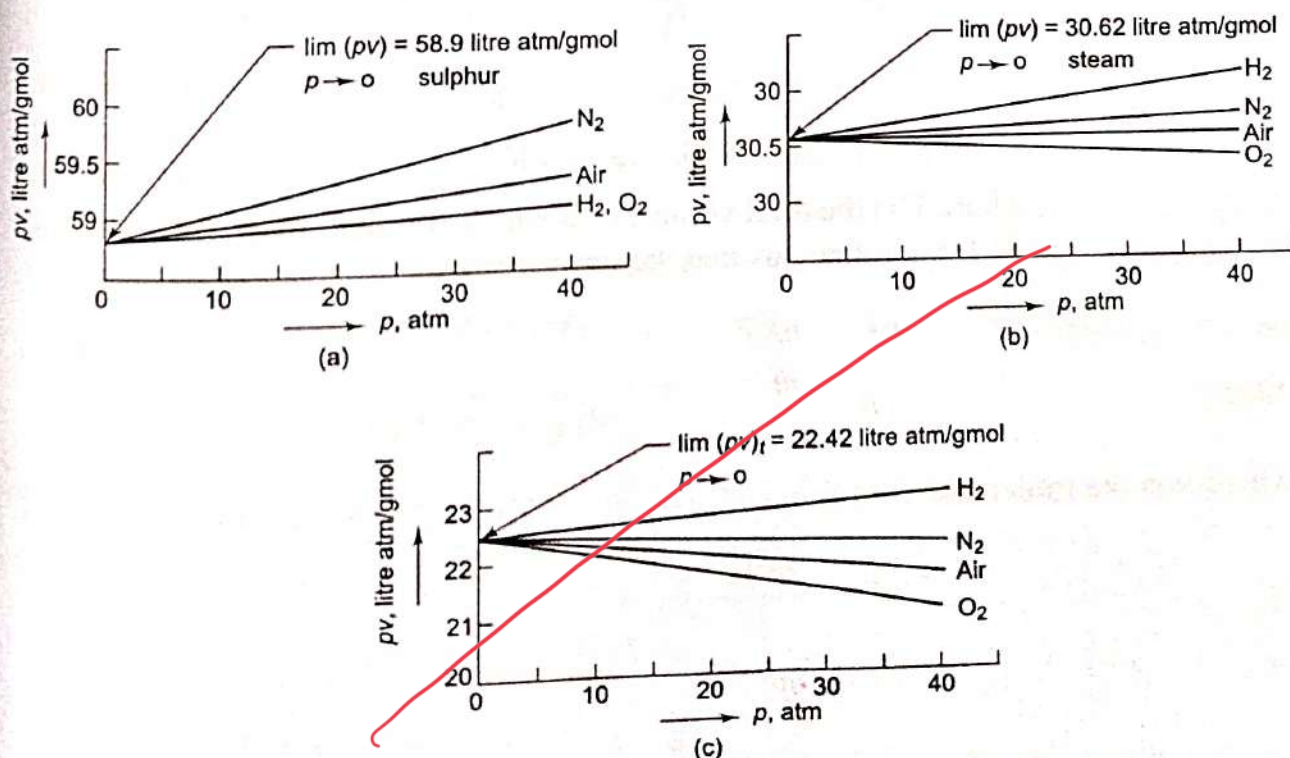
The ideal gas temperature  $T$ , is thus

$$T = 273.16 \frac{\lim_{p \rightarrow 0} (pv)}{\lim_{p \rightarrow 0} (pv)_t}$$

$$\therefore \lim_{p \rightarrow 0} (pv) = \left[ \frac{\lim_{p \rightarrow 0} (pv)_t}{273.16} \right] T$$

The term within bracket is called the *universal gas constant* and is denoted by  $\bar{R}$ . Thus,

$$\bar{R} = \frac{\lim_{p \rightarrow 0} (pv)_t}{273.16} \quad (10.2)$$



**Fig. 10.1** For any gas  $\lim_{p \rightarrow 0} (pv)_T$  is independent of the nature of the gas and depends only on  $T$

The value obtained for  $\lim_{p \rightarrow 0} (pv)_t$  is  $22.4 \frac{\text{litre} - \text{atm}}{\text{g mol}}$

$$\therefore \bar{R} = \frac{22.4}{273.16} = 0.083 \frac{\text{litre} - \text{atm}}{\text{g mol K}}$$

The equation of state of a gas is thus

$$\lim_{p \rightarrow 0} p\bar{v} = \bar{R}T \quad (10.3)$$

where  $\bar{v}$  is the molar volume.

### 10.3 IDEAL GAS

A hypothetical gas which obeys the law  $p\bar{v} = \bar{R}T$  at all pressures and temperatures is called an *ideal gas*.

Real gases do not conform to this equation of state with complete accuracy. As  $p \rightarrow 0$ , or  $T \rightarrow \infty$ , the real gas approaches the ideal gas behaviour. In the equation,  $p\bar{v} = \bar{R}T$ , as  $T \rightarrow 0$ , i.e.

$t \rightarrow -273.15^\circ\text{C}$ , if  $\bar{v}$  remains constant,  $p \rightarrow 0$ , or if  $p$  remains constant  $\bar{v} \rightarrow 0$ . Since negative volume or negative pressure is inconceivable, the lowest possible temperature is 0 K or  $-273.15^\circ\text{C}$ .  $T$  is, therefore, known as the *absolute temperature*.

There is no permanent or perfect gas. At atmospheric condition only, these gases exist in the gaseous state. They are subject to liquefaction or solidification, as the temperature and pressure are sufficiently lowered.

From Avogadro's law, when  $p = 760 \text{ mm Hg} = 1.013 \times 10^5 \text{ N/m}^2$ ,  $T = 273.15 \text{ K}$ , and  $\bar{v} = 22.4 \text{ m}^3/\text{kg mol}$

$$\begin{aligned} \bar{R} &= \frac{1.013 \times 10^5 \times 22.4}{273.15} \\ &= 8314.3 \text{ Nm/kg mol K} \\ &= 8.3143 \text{ kJ/kg mol K} \end{aligned}$$

Since  $\bar{v} = V/n$ , where  $V$  is the total volume and  $n$  is the number of moles of the gas, the equation of state for an ideal gas may be written as

$$pV = n\bar{R}T \quad (10.4)$$

Also

$$n = \frac{m}{\mu}$$

where  $\mu$  is the molecular weight

$$\therefore pV = \frac{m\bar{R}T}{\mu}$$

or

$$pV = mRT \quad (10.5)$$

where  $R = \text{characteristic gas constant} = \frac{\bar{R}}{\mu}$  (10.6)

#### LO 10.3

Summarize various equations of state of an ideal gas



For oxygen, e.g.

$$R_{O_2} = \frac{8.3143}{32} = 0.262 \text{ kJ/kg K}$$

For air,

$$R_{\text{air}} = \frac{8.3143}{28.96} = 0.287 \text{ kJ/kg K}$$

There are  $6.023 \times 10^{23}$  molecules in a g mol of a substance. This is known as *Avogadro's number* ( $A$ ).

$\therefore A = 6.023 \times 10^{26}$  molecules/kg mol  
In  $n$  kg moles of gas, the total number of molecules,  $N$ , are

$$N = nA$$

or

$$n = N/A$$

$$pV = N \frac{\bar{R}}{A} T$$

$$= NKT$$

(10.7)

where  $K$  = Boltzmann constant

$$= \frac{\bar{R}}{A} = \frac{8314.3}{6.023 \times 10^{26}} = 1.38 \times 10^{-23} \text{ J/molecule K}$$

Therefore, the equation of state of an ideal gas is given by

$$\begin{aligned} pV &= mRT \\ &= n\bar{R}T \\ &= NKT \end{aligned}$$

### 10.3.1 Specific Heats, Internal Energy and Enthalpy of an Ideal Gas

An ideal gas not only satisfies the equation of state  $pv = RT$ , but its *specific heats are constant* also. For real gases, these vary appreciably with temperature, and little with pressure.

The properties of a substance are related by

*No Derivation*  $Tds = du + pdv$

or  $ds = \frac{du}{T} + \frac{p}{T} dv$

(10.8)

The internal energy  $u$  is assumed to be a function of  $T$  and  $v$ , i.e.

$$u = f(T, v)$$

or

$$du = \left( \frac{\partial u}{\partial T} \right)_v dT + \left( \frac{\partial u}{\partial v} \right)_T dv$$

(10.9)

From Eqs. (10.8) and (10.9)

$$ds = \frac{1}{T} \left( \frac{\partial u}{\partial T} \right)_v dT + \frac{1}{T} \left[ \left( \frac{\partial u}{\partial v} \right)_T + p \right] dv$$

(10.10)

#### LO 10.4

Relate specific heat with internal energy and enthalpy of an ideal gas to derive an expression of Gibb's function of inert ideal gases mixture

Again, let

$$s = f(T, v)$$

$$ds = \left( \frac{\partial s}{\partial T} \right)_v dT + \left( \frac{\partial s}{\partial v} \right)_T dv \quad (10.11)$$

Comparing Eqs. (10.10) and (10.11)

$$\left( \frac{\partial s}{\partial T} \right)_v = \frac{1}{T} \left( \frac{\partial u}{\partial T} \right)_v \quad (10.12)$$

$$\left( \frac{\partial s}{\partial v} \right)_T = \frac{1}{T} \left[ \left( \frac{\partial u}{\partial v} \right)_T + p \right] \quad (10.13)$$

Differentiating Eq. (10.12) with respect to  $v$  when  $T$  is constant

$$\frac{\partial^2 s}{\partial T \partial v} = \frac{1}{T} \frac{\partial^2 u}{\partial T \partial v} \quad (10.14)$$

Differentiating Eq. (10.13) with respect to  $T$  when  $v$  is constant

$$\frac{\partial^2 s}{\partial v \partial T} = \frac{1}{T} \frac{\partial^2 u}{\partial v \partial T} + \frac{1}{T} \left( \frac{\partial p}{\partial T} \right)_v - \frac{1}{T^2} \left( \frac{\partial u}{\partial v} \right)_T - \frac{p}{T^2} \quad (10.15)$$

From Eqs. (10.14) and (10.15)

$$\frac{1}{T} \frac{\partial^2 u}{\partial T \partial v} = \frac{1}{T} \frac{\partial^2 u}{\partial v \partial T} + \frac{1}{T} \left( \frac{\partial p}{\partial T} \right)_v - \frac{1}{T^2} \left( \frac{\partial u}{\partial v} \right)_T - \frac{p}{T^2}$$

or

$$\left( \frac{\partial u}{\partial v} \right)_T + p = T \left( \frac{\partial p}{\partial T} \right)_v \quad (10.16)$$

For an ideal gas

$$pv = RT$$

$$v \left( \frac{\partial p}{\partial T} \right)_v = R$$

$$\left( \frac{\partial p}{\partial T} \right)_v = \frac{R}{v} = \frac{p}{T} \quad (10.17)$$

From Eqs. (10.16) and (10.17)

$$\left( \frac{\partial u}{\partial v} \right)_T = 0 \quad (10.18)$$

Therefore,  $u$  does not change when  $v$  changes at constant temperature. Similarly, if  $u = f(T, p)$ , it can be shown that  $\left( \frac{\partial u}{\partial p} \right)_T = 0$ . Therefore,  $u$  does not change with  $p$  either, when  $T$  remains constant.

$u$  does not change unless  $T$  changes.

Then

$$u = f(T)$$

(10.19)

only remember this relation



only for an ideal gas. This is known as *Joule's law*.

If

$$u = f(T, v)$$

$$du = \left( \frac{\partial u}{\partial T} \right)_v dT + \left( \frac{\partial u}{\partial v} \right)_T dv$$

Since the last term is zero by Eq. (10.18), and by definition

$$c_v = \left( \frac{\partial u}{\partial T} \right)_v$$

$$du = c_v dT$$

(10.20)

The equation  $du = c_v dT$  holds good for an ideal gas for any process, whereas for any other substance it is true for a constant volume process only.

Since  $c_v$  is constant for an ideal gas,

$$\Delta u = c_v \Delta T$$

The enthalpy of any substance is given by

$$h = u + pv$$

For an ideal gas

$$h = u + RT$$

Therefore,

$$h = f(T)$$

(10.21)

only for an ideal gas.

Now

$$dh = du + R dT$$

Since  $R$  is a constant

$$\begin{aligned} \Delta h &= \Delta u + R \Delta T \\ &= c_v \Delta T + R \Delta T \\ &= (c_v + R) \Delta T \end{aligned}$$

(10.22)

Since  $h$  is a function of  $T$  only, and by definition

$$c_p = \left( \frac{\partial h}{\partial T} \right)_p$$

$$dh = c_p dT$$

(10.23)

or

$$\Delta h = c_p \Delta T$$

(10.24)

From Eqs. (10.22) and (10.23)

$$c_p = c_v + R$$

or

$$c_p - c_v = R$$

(10.25)

The equation  $dh = c_p dT$  holds good for an ideal gas, even when pressure changes, but for any other substance, this is true only for a constant pressure change.

The ratio of  $c_p/c_v$  is of importance in ideal gas computations and is designated by the symbol  $\gamma$ , i.e.

$$\frac{c_p}{c_v} = \gamma$$

$$c_p = \gamma c_v$$

or

From Eq. (10.25)

$$(\gamma - 1) c_v = R$$

$$\text{and } \left. \begin{aligned} c_v &= \frac{R}{\gamma - 1} \\ c_p &= \frac{\gamma R}{\gamma - 1} \end{aligned} \right\} \text{kJ/kg K} \quad (10.26)$$

If  $R = \frac{\bar{R}}{\mu}$  is substituted in Eq. (10.26)

$$\text{and } \left. \begin{aligned} \bar{c}_v &= \mu c_v = (c_v)_{\text{mol}} = \frac{\bar{R}}{\gamma - 1} \\ \bar{c}_p &= \mu c_p = (c_p)_{\text{mol}} = \frac{\gamma \bar{R}}{\gamma - 1} \end{aligned} \right\} \text{kJ/(kg mol)(K)} \quad (10.27)$$

$\bar{c}_v$  and  $\bar{c}_p$  are the *molar or molal specific heats* at constant volume and at constant pressure, respectively.

It can be shown by the classical kinetic theory of gases that the values of  $\gamma$  are 5/3 for monatomic gases and 7/5 for diatomic gases. When the gas molecule contains more than two atoms (i.e. for polyatomic gases) the value of  $\gamma$  may be taken approximately as 4/3. The minimum value of  $\gamma$  is thus 1 and the maximum is 1.67.

The value of  $\gamma$  thus depends only on the molecular structure of the gas, i.e. whether the gas is monatomic, diatomic or polyatomic having one, two or more atoms in a molecule. It may be noted that  $c_p$  and  $c_v$  of an ideal gas depend only on  $\gamma$  and  $R$ , i.e. the number of atoms in a molecule and the molecular weight of the gas. They are independent of temperature or pressure of the gas.

**Further considerations** An important ratio that we shall be using extensively is that of the specific heats,

$$\gamma = \frac{c_p}{c_v}$$

which has so far been defined on a mass basis. It can also be written on the mole basis. We denote the molal constant volume specific heat as

$$\bar{c}_v = \left( \frac{\partial \bar{u}}{\partial T} \right)_v \text{ kJ/kgmol-K}$$

and the molal constant pressure specific heat as

$$\bar{c}_p = \left( \frac{\partial \bar{h}}{\partial T} \right)_p \text{ kJ/kgmol-K}$$

The bar over any property denotes that it is on the mole basis; hence, both enthalpy  $\bar{h}$  and internal energy  $\bar{u}$  have units of kJ/kgmol. The property on the mass basis is multiplied by the molecular weight of the substance,  $\mu$ , to obtain the property on the mole bases, e.g.  $\bar{h} = \mu h$ .

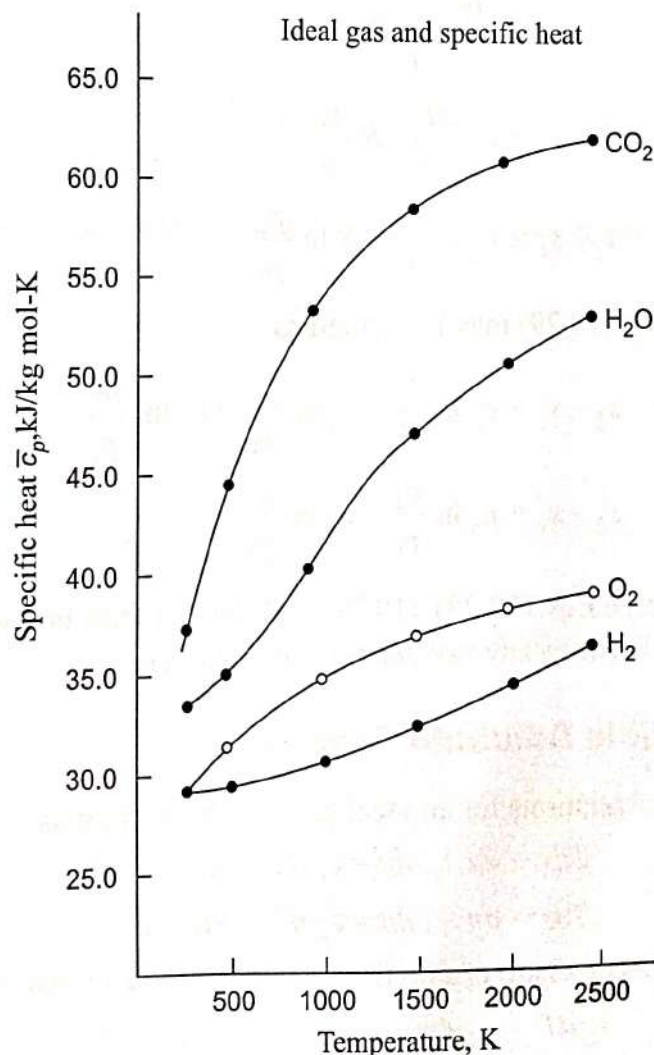
**Variation of specific heat with temperature** The specific heat of a real gas varies with temperature. The functional relationships denoting this variation are determined from experimental tests. Table 10.1 lists several formulas that predict the specific heat of a gas at a given temperature.



Figure 10.2 shows a diagram of the specific heat variation with temperature for a few gases.

**Table 10.1** Formulas for specific heat variation with temperature

| Substance and temperature range                                | $c_p$ (kJ/kgK)                                                |
|----------------------------------------------------------------|---------------------------------------------------------------|
| Air, 280–1500 K                                                | $0.9167 + 2.577 \times 10^{-4}T - 3.974 \times 10^{-8}T^2$    |
| Ammonia, 300–1000 K                                            | $1.5194 + 1.936 \times 10^{-3}T - 1.789 \times 10^{-7}T^2$    |
| Sulphur dioxide, 300–1000 K                                    | $0.7848 + 0.7113 \times 10^{-4}T - 1.73 \times 10^{-4}T^{-2}$ |
| Hydrogen, 300–2200 K                                           | $11.959 + 2.160 \times 10^{-3}T + 30.95T^{-1/2}$              |
| Oxygen, 300–2750 K                                             | $1.507 - 16.77T^{-1/2} + 111.1T^{-1}$                         |
| Nitrogen, 300–5000 K                                           | $1.415 - 287.9T^{-1} + 5.35 \times 10^{-4}T^{-2}$             |
| Carbon monoxide, 300–5000 K                                    | $1.415 - 273.3T^{-1} + 4.96 \times 10^{-4}T^{-2}$             |
| H <sub>2</sub> O, 300–3000 K                                   | $4.613 - 103.3T^{-1/2} + 967.5T^{-1}$                         |
| Carbon dioxide, 300–3500 K                                     | $1.540 - 345.1T^{-1} + 4.13 \times 10^{-4}T^{-2}$             |
| Methane, 300–1500 K                                            | $0.8832 + 4.71 \times 10^{-3}T - 1.123 \times 10^{-6}T^2$     |
| Ethylene, 300–1500 K                                           | $0.4039 + 4.35 \times 10^{-3}T - 1.35 \times 10^{-6}T^2$      |
| Ethane, 300–1500 K                                             | $0.306 + 5.34 \times 10^{-3}T - 1.53 \times 10^{-6}T^2$       |
| <i>n</i> -butane (C <sub>4</sub> H <sub>10</sub> ), 300–1500 K | $0.314 + 5.23 \times 10^{-3}T - 1.60 \times 10^{-6}T^2$       |
| Propane, 300–1500 K                                            | $0.214 + 5.48 \times 10^{-3}T - 1.67 \times 10^{-6}T^2$       |
| Acetylene (C <sub>2</sub> H <sub>2</sub> ), 280–1250 K         | $1.921 + 7.06 \times 10^{-4}T - 3.73 \times 10^{-4}T^{-2}$    |
| Octane, 225–610 K                                              | $0.290 + 3.97 \times 10^{-3}T$                                |



**Fig. 10.2** Variation of  $c_p$  with temperature for four gases, plotted from equations in Table 10.1

**Gas tables** Table A-12 in Appendix A provides the properties of air which account for the variation of specific heat with temperature for an ideal gas. Temperature is the independent variable and values of internal energy, enthalpy, etc. are given on the same line.

### 10.3.2 Entropy Change of an Ideal Gas

From the general property relations

$$Tds = du + pdv$$

$$Tds = dh - vdp$$

and for an ideal gas,  $du = c_v dT$ ,  $dh = c_p dT$  and  $pv = RT$ , the entropy change between any two states 1 and 2 may be computed as given below.

$$ds = \frac{du}{T} + \frac{p}{T} dv$$

$$= c_v \frac{dT}{T} + R \frac{dv}{v}$$

$\therefore$

$$s_2 - s_1 = c_v \ln \frac{T_2}{T_1} + R \ln \frac{v_2}{v_1} \quad (10.28)$$

Also

$$ds = \frac{dh}{T} - \frac{v}{T} dp$$

$$= c_p \frac{dT}{T} - R \frac{dp}{p}$$

or

$$s_2 - s_1 = c_p \ln \frac{T_2}{T_1} - R \ln \frac{p_2}{p_1} \quad (10.29)$$

Since  $R = c_p - c_v$ , Eq. (10.29) may be written as

$$s_2 - s_1 = c_p \ln \frac{T_2}{T_1} - c_p \ln \frac{p_2}{p_1} + c_v \ln \frac{p_2}{p_1}$$

or

$$s_2 - s_1 = c_p \ln \frac{v_2}{v_1} + c_v \ln \frac{p_2}{p_1} \quad (10.30)$$

Any one of the three Eqs. (10.28), (10.29) and (10.30), may be used for computing the entropy change between any two states of an ideal gas.

### 10.3.3 Reversible Adiabatic Process

The general property relations for an ideal gas may be written as

$$Tds = du + pdv = c_v dT + pdv$$

and

$$Tds = dh - vdp = c_p dT - vdp$$

For a reversible adiabatic change,  $ds = 0$

$\therefore$

$$c_v dT = -pdv$$

(10.31)

LO 10.5

Derive entropy change of an ideal gas and a relation between  $p$ ,  $v$  and  $T$  in a reversible adiabatic process

$$\left( \frac{p_2}{p_1} \right)^{\gamma-1} = \frac{T_2}{T_1} = \left( \frac{v_1}{v_2} \right)^{\gamma-1}$$



and

By division

$$c_p dT = v dp \quad (10.32)$$

$$\frac{c_p}{c_v} = \gamma = - \frac{v dp}{p dv}$$

or

$$\frac{dp}{p} + \gamma \frac{dv}{v} = 0$$

or

$$d(\ln p) + \gamma d(\ln v) = d(\ln c)$$

where  $c$  is a constant.

$\therefore$

$$\ln p + \gamma \ln v = \ln c$$

$$pv^\gamma = c$$

(10.33)

Between any two states 1 and 2

$$p_1 v_1^\gamma = p_2 v_2^\gamma$$

or

$$\frac{p_2}{p_1} = \left( \frac{v_1}{v_2} \right)^\gamma$$

For an ideal gas

$$pv = RT$$

From Eq. (10.33)

$$p = c \cdot v^{-\gamma}$$

$\therefore$

$$c \cdot v^{-\gamma} \cdot v = RT$$

$$c \cdot v^{1-\gamma} = RT$$

$$Tv^{\gamma-1} = \text{constant}$$

(10.34)

Between any two states 1 and 2, for a reversible adiabatic process in the case of an ideal gas

$$T_1 v_1^{\gamma-1} = T_2 v_2^{\gamma-1}$$

or

$$\frac{T_2}{T_1} = \left( \frac{v_1}{v_2} \right)^{\gamma-1}$$

(10.35)

Similarly, substituting from Eq. (10.33)

$$v = \left( \frac{c}{p} \right)^{1/\gamma} \text{ in the Eq. } pv = RT$$

$$p \cdot \frac{C'}{p^{1/\gamma}} = RT$$

$$p^{1-(1/\gamma)} \cdot c' = RT$$

$\therefore$

$$Tp^{(1-\gamma)/\gamma} = \text{constant}$$

(10.36)

Between any two states 1 and 2

$$T_1 p_1^{(1-\gamma)/\gamma} = T_2 p_2^{(1-\gamma)/\gamma}$$

$$\therefore \frac{T_2}{T_1} = \left( \frac{p_2}{p_1} \right)^{(\gamma-1)/\gamma} \quad (10.37)$$

Equations (10.33), (10.34) and (10.36) give the relations among  $p$ ,  $v$  and  $T$  in a reversible adiabatic process for an ideal gas.

The *internal energy change* of an ideal gas for a reversible adiabatic process is given by

$$Tds = du + pdv = 0$$

or

$$\int_1^2 du = -\int_1^2 pdv = -\int_1^2 \frac{c}{v^\gamma} dv$$

where

$$pv^\gamma = p_1 v_1^\gamma = p_2 v_2^\gamma = c$$

$\therefore$

$$\begin{aligned} u_2 - u_1 &= c \frac{v_2^{1-\gamma} - v_1^{1-\gamma}}{\gamma-1} = \frac{p_2 v_2^\gamma \cdot v_2^{1-\gamma} - p_1 v_1^\gamma \cdot v_1^{1-\gamma}}{\gamma-1} \\ &= \frac{p_2 v_2 - p_1 v_1}{\gamma-1} \\ &= \frac{R(T_2 - T_1)}{\gamma-1} = \frac{RT_1}{\gamma-1} \left( \frac{T_2}{T_1} - 1 \right) \\ &= \frac{RT_1}{\gamma-1} \left[ \left( \frac{p_2}{p_1} \right)^{\gamma-1/\gamma} - 1 \right] \end{aligned} \quad (10.38)$$

The *enthalpy change* of an ideal gas for a reversible adiabatic process may be similarly derived.

$$Tds = dh - vdp = 0$$

or

$$\int_1^2 dh = \int_1^2 vdp = \int_1^2 \frac{(c)^{1/\gamma}}{p^{1/\gamma}} vdp$$

where

$$p_1 v_1^\gamma = p_2 v_2^\gamma = c$$

$\therefore$

$$\begin{aligned} h_2 - h_1 &= \frac{\gamma}{\gamma-1} c^{1/\gamma} [p_2^{(\gamma-1)/\gamma} - p_1^{(\gamma-1)/\gamma}] \\ &= \frac{\gamma}{\gamma-1} (p_1 v_1^\gamma)^{1/\gamma} \cdot (p_1)^{(\gamma-1)/\gamma} \left[ \left( \frac{p_2}{p_1} \right)^{(\gamma-1)/\gamma} - 1 \right] \\ &= \frac{\gamma p_1 v_1}{\gamma-1} \left[ \left( \frac{p_2}{p_1} \right)^{(\gamma-1)/\gamma} - 1 \right] \\ &= \frac{\gamma RT_1}{\gamma-1} \left[ \left( \frac{p_2}{p_1} \right)^{(\gamma-1)/\gamma} - 1 \right] \end{aligned} \quad (10.39)$$



The work done by an ideal gas in a reversible adiabatic process is given by

$$\delta Q = dU + \delta W = 0$$

or  $\delta W = -dU$

i.e. work is done at the expense of the internal energy.

$$\therefore W_{1-2} = U_1 - U_2 = m(u_1 - u_2)$$

$$= \frac{m(p_1 v_1 - p_2 v_2)}{\gamma - 1} = \frac{mR(T_1 - T_2)}{\gamma - 1} = \frac{mRT_1}{\gamma - 1} \left[ 1 - \left( \frac{p_2}{p_1} \right)^{(\gamma-1)/\gamma} \right] \quad (10.40)$$

where  $m$  is the mass of gas.

In a *steady flow process*, where both flow work and external work are involved, we have from SFEE,

$$W_x + \Delta \frac{V^2}{2} + g\Delta z = h_1 - h_2 = c_p(T_1 - T_2) = \frac{\gamma R(T_1 - T_2)}{\gamma - 1}$$

$$= \frac{\gamma(p_1 v_1 - p_2 v_2)}{\gamma - 1} = \frac{\gamma}{\gamma - 1} p_1 v_1 \left[ 1 - \left( \frac{p_2}{p_1} \right)^{\gamma-1/\gamma} \right] \quad (10.41)$$

If K.E. and P.E. changes are neglected,

$$W_x = \frac{\gamma}{\gamma - 1} p_1 v_1 \left[ 1 - \left( \frac{p_2}{p_1} \right)^{\gamma-1/\gamma} \right] \quad (10.42)$$

### 10.3.4 Reversible Isothermal Process

When an ideal gas of mass  $m$  undergoes a reversible isothermal process from state 1 to 2, the work done is given by

$$\int_1^2 \delta W = \int_{V_1}^{V_2} p dV$$

or

$$W_{1-2} = \int_{V_1}^{V_2} \frac{mRT}{V} dV = mRT \ln \frac{V_2}{V_1}$$

$$= mRT \ln \frac{p_1}{p_2} \quad (10.43)$$

The heat transfer involved in the process

$$Q_{1-2} = U_2 - U_1 + W_{1-2}$$

$$= W_{1-2} = mRT \ln V_2/V_1 = T(S_2 - S_1) \quad (10.44)$$

### 10.3.5 Polytropic Process

An equation of the form  $pv^n = \text{constant}$ , where  $n$  is a constant can be used approximately to describe many processes which occur in practise. Such a process is called *polytropic process*. It is not adiabatic, but it can be reversible.

#### LO 10.6

Derive an expression to calculate work transfer in a reversible isothermal process

#### LO 10.7

Define polytropic process and discuss its significance

It may be noted that  $\gamma$  is a property of the gas, whereas  $n$  is not. The value of  $n$  depends upon the process. It is possible to find the value of  $n$  which more or less fits the experimental results. For two states on the process,

$$p_1 v_1^n = p_2 v_2^n \quad (10.45)$$

or

$$\left(\frac{v_2}{v_1}\right)^n = \frac{p_1}{p_2}$$

$$n = \frac{\log p_1 - \log p_2}{\log v_2 - \log v_1} \quad (10.46)$$

For known values of  $p_1, p_2, v_1$  and  $v_2$ ,  $n$  can be estimated from the above relation.

Two other relations of a polytropic process, corresponding to Eqs. (10.35) and (10.37) are:

$$\frac{T_2}{T_1} = \left(\frac{v_1}{v_2}\right)^{n-1} \quad (10.47)$$

$$\frac{T_2}{T_1} = \left(\frac{p_2}{p_1}\right)^{n-1/n} \quad (10.48)$$

**1. Entropy change in a polytropic process** In a reversible adiabatic process, the entropy remains constant. But in a reversible polytropic process, the entropy changes. Substituting Eqs. (10.45), (10.47) and (10.48) in Eqs. (10.28), (10.29) and (10.30) respectively, we have three expressions for entropy change given as follows:

$$\begin{aligned} s_2 - s_1 &= c_v \ln \frac{T_2}{T_1} + R \ln \frac{v_2}{v_1} \\ &= \frac{R}{\gamma-1} \ln \frac{T_2}{T_1} + \frac{R}{n-1} \ln \frac{T_1}{T_2} \\ &= \frac{n-\gamma}{(\gamma-1)(n-1)} R \ln \frac{T_2}{T_1} \end{aligned} \quad (10.49)$$

Relations in terms of pressure and specific volume can be similarly derived. These are

$$s_2 - s_1 = \frac{n-\gamma}{n(\gamma-1)} R \ln \frac{p_2}{p_1} \quad (10.50)$$

and

$$s_2 - s_1 = -\frac{n-\gamma}{\gamma-1} R \ln \frac{v_2}{v_1} \quad (10.51)$$

It can be noted that when  $n = \gamma$ , the entropy change becomes zero. If  $p_2 > p_1$ , for  $n \leq \gamma$ , the entropy of the gas decreases, and for  $n > \gamma$ , the entropy of the gas increases.



The increase of entropy may result from reversible heat transfer to the system from the surroundings. Entropy decrease is also possible if the gas is cooled.

**2. Heat and work in a polytropic process** Using the first law to unit mass of an ideal gas,

$$\begin{aligned}
 Q - W &= u_2 - u_1 \\
 &= c_v (T_2 - T_1) = \frac{R(T_2 - T_1)}{\gamma - 1} \\
 &= \frac{p_2 v_2 - p_1 v_1}{\gamma - 1} \\
 &= \frac{p_1 v_1}{\gamma - 1} \left[ \left( \frac{p_2}{p_1} \right)^{n-1/n} - 1 \right] = \frac{p_1 v_1}{\gamma - 1} \left[ \left( \frac{v_1}{v_2} \right)^{n-1} - 1 \right] \quad (10.52)
 \end{aligned}$$

For a steady flow system of unit mass of ideal gas, the SFEE Eq. (5.10) gives

$$\begin{aligned}
 Q - W_x - \Delta \left[ \frac{\bar{V}_2^2}{2} + gz \right] &= h_2 - h_1 \\
 &= c_p (T_2 - T_1) = \frac{\gamma R (T_2 - T_1)}{\gamma - 1} \\
 &= \frac{\gamma}{\gamma - 1} (p_2 v_2 - p_1 v_1) \quad (10.53)
 \end{aligned}$$

For a polytropic process,

$$\begin{aligned}
 Q - W_x - \Delta \left[ \frac{\bar{V}_2^2}{2} + gz \right] &= \frac{\gamma p_1 v_1}{\gamma - 1} \left[ \left( \frac{p_2}{p_1} \right)^{n-1/n} - 1 \right] \\
 &= \frac{\gamma p_1 v_1}{\gamma - 1} \left[ \left( \frac{v_1}{v_2} \right)^{n-1} - 1 \right] \quad (10.54)
 \end{aligned}$$

Equations (10.52) and (10.54) can be used to determine heat and work quantities for a closed and a steady flow system, respectively.

**3. Integral property relations in a polytropic process** In a  $p v^n = \text{constant}$  process,

$$\begin{aligned}
 \int_1^2 p dv &= \int_1^2 \frac{p_1 v_1^n}{v^n} dv = \frac{p_1 v_1}{n-1} \left[ 1 - \left( \frac{v_1}{v_2} \right)^{n-1} \right] \\
 &= \frac{p_1 v_1}{n-1} \left[ 1 - \left( \frac{p_2}{p_1} \right)^{n-1/n} \right] \quad (10.55)
 \end{aligned}$$

Similarly,

$$\begin{aligned} -\int_1^2 v dp &= \frac{np_1 v_1}{n-1} \left[ 1 - \left( \frac{v_1}{v_2} \right)^{n-1} \right] \\ &= \frac{np_1 v_1}{n-1} \left[ 1 - \left( \frac{p_2}{p_1} \right)^{n-1/n} \right] \end{aligned} \quad (10.56)$$

The integral of  $Tds$  is obtained from the property relation

$$\begin{aligned} Tds &= du + pdv \\ \therefore \int_1^2 Tds &= \int_1^2 du + \int_1^2 pdv \\ &= u_2 - u_1 + \int_1^2 pdv \end{aligned}$$

Substituting from Eqs. (10.50) and (10.53)

$$\begin{aligned} \int_1^2 Tds &= \frac{\gamma - n}{(\gamma - 1)(n - 1)} p_1 v_1 \left[ 1 - \left( \frac{p_2}{p_1} \right)^{n-1/n} \right] \\ &= \frac{\gamma - n}{(\gamma - 1)(n - 1)} p_1 v_1 \left[ 1 - \left( \frac{v_1}{v_2} \right)^{n-1} \right] \\ &= \frac{\gamma - n}{(\gamma - 1)(n - 1)} R(T_1 - T_2) \end{aligned} \quad (10.57)$$

Since  $R/(\gamma - 1) = c_v$ , and putting  $\Delta T = T_2 - T_1$ , the reversible heat transfer

$$\begin{aligned} Q_R &= \int_1^2 Tds = c_v \frac{\gamma - n}{1 - n} \Delta T \\ &= c_n \Delta T \end{aligned} \quad (10.58)$$

where  $c_n = c_v(\gamma - n)/(1 - n)$  is called the *polytropic specific heat*. For  $n > \gamma$ , there will be positive heat transfer and gain in entropy. For  $n < \gamma$ , heat transfer will be negative and entropy of the gas would decrease.

Equation (10.58) can also be derived in the following way. For a polytropic process,

$$\begin{aligned} dQ &= du + pdv \\ Q &= c_v(T_2 - T_1) + \frac{R(T_1 - T_2)}{n - 1} = c_v(T_2 - T_1) - \frac{c_v(\gamma - 1)(T_2 - T_1)}{n - 1} \\ &= c_v(T_2 - T_1) \left[ 1 - \frac{\gamma - 1}{n - 1} \right] = c_v \frac{n - \gamma}{n - 1} (T_2 - T_1) \\ &= c_n(T_2 - T_1) \end{aligned}$$

Ordinarily, both heat and work are involved in a polytropic process. To evaluate the heat transfer during such a process, it is necessary to first evaluate the work via



either  $\int p dv$  or  $-\int v dp$ , depending on whether it is a closed or an open steady flow system. The application of the first law will then yield the heat transfer.

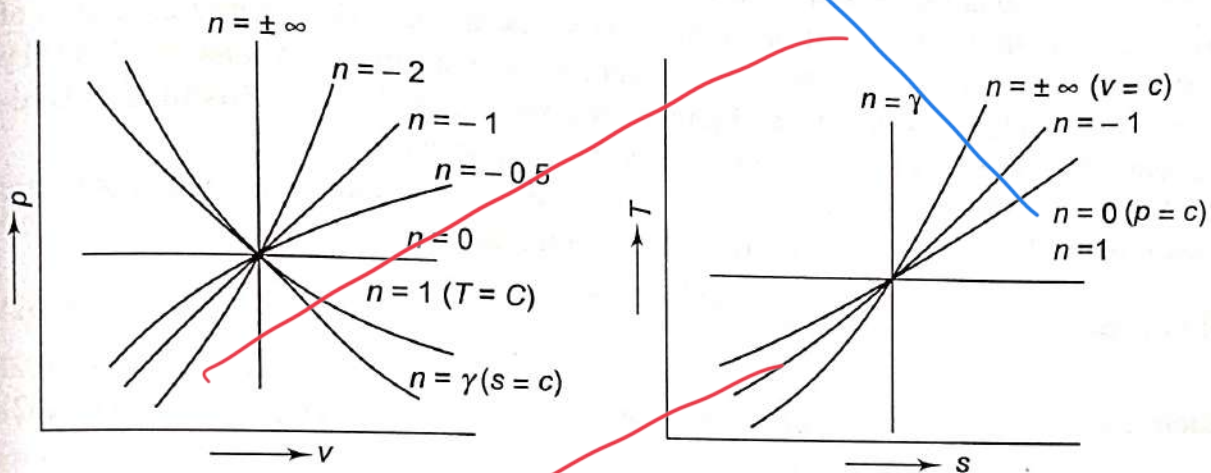
The polytropic processes for various values of  $n$  are shown in Fig. 10.3 on the  $p$ - $v$  and  $T$ - $s$  diagrams.

On differentiation,

$$pv^n = C$$

$$v^n dp + pn v^{n-1} dv = 0$$

$$\frac{dp}{dv} = -n \frac{p}{v} \quad (10.59)$$



**Fig. 10.3** Process in which  $pv^n = \text{Constant}$

The slope of the curve increases in the negative direction with increase of  $n$ . The values of  $n$  for some familiar processes are given below.

Isobaric process ( $p = c$ ),  $n = 0$

Isothermal process ( $T = c$ ),  $n = 1$

Isentropic process ( $s = c$ ),  $n = \gamma$

Isometric or isochoric process ( $v = c$ ),  $n = \infty$ .

## 10.4 EQUATIONS OF STATE

The ideal gas equation of state  $p\bar{v} = \bar{R}T$  can be established from the postulates of the kinetic theory of gases developed by Clerk Maxwell, with two important assumptions that there is little or no attraction between the molecules of the gas and that the volume occupied by the molecules themselves is negligibly small compared to the volume of the gas. When pressure is very small or temperature is very large, the intermolecular attraction and the volume of the molecules compared to the total volume of the gas are not of much significance, and the real gas obeys very closely the ideal gas equation. But as pressure increases, the intermolecular forces of attraction and repulsion increase, and also the volume of the molecules becomes appreciable compared to the total gas volume. So then the real gases deviate considerably from the ideal gas equation.

### LO 10.8

Explain the equations of state proposed for real gases



van der Waals, by applying the laws of mechanics to individual molecules, introduced two correction terms in the equation of ideal gas, and his equation is given below.

$$\left(p + \frac{a}{v^2}\right)(v - b) = RT \quad (10.60)$$

The coefficient  $a$  was introduced to account for the existence of mutual attraction between the molecules. The term  $a/v^2$  is called the *force of cohesion*. The coefficient  $b$  was introduced to account for the volumes of the molecules, and is known as *co-volume*.

Real gases conform more closely with the van der Waals equation of state than the ideal gas equation of state, particularly at higher pressures. But it is not obeyed by a real gas in all ranges of pressures and temperatures. Many more equations of state were later introduced, and notable among these are the equations developed by Berthelot, Dieterici, Beattie-Bridgeman, Kammerlingh Onnes, Hirshfelder-Bird-Sportz-McGee-Sutton, Wohl Redlich-Kwong and Martin-Hou.

Apart from the van der Waals equation, three two-constant equations of state are those of Berthelot, Dieterici and Redlich-Kwong, as given below:

Berthelot: 
$$p = \frac{RT}{v - b} - \frac{a}{Tv^2} \quad (10.61)$$

Dieterici: 
$$p = \frac{RT}{v - b} \cdot e^{-a/RTv} \quad (10.62)$$

Redlich-Kwong: 
$$p = \frac{RT}{v - b} - \frac{a}{T^{1/2}v(v + b)} \quad (10.63)$$

The constants,  $a$  and  $b$  are evaluated from the critical data, as shown for van der Waals equation in Section 10.7. The Berthelot and Dieterici equations of state, like the van der Waals equation, are of limited accuracy. But the Redlich-Kwong equation gives good results at high pressures and is fairly accurate for temperatures above the critical value.

Another two-constant equation which is again of limited accuracy is the Saha-Bose equation of state as given below.

$$p = -\frac{RT}{2b} e^{-a/RTv} \ln \left( \frac{v - 2b}{v} \right) \quad (10.64)$$

It is, however, quite accurate for densities less than about 0.8 times the critical density.

One more widely used equation of state with good accuracy is the Beattie-Bridgeman equation:

$$p = \frac{RT(1 - e)}{v^2} (v + B) - \frac{A}{v^2} \quad (10.65)$$

where

$$A = A_0 \left( 1 - \frac{a}{v} \right), B = B_0 \left( 1 - \frac{b}{v} \right), e = \frac{c}{vT^3}$$

There are five constants,  $A_0$ ,  $B_0$ ,  $a$ ,  $b$  and  $c$ , which have to be determined experimentally for each gas. The Beattie-Bridgeman equation does not give satisfactory results in the critical point region.



All these equations mentioned above reduce to the ideal gas equation for large volumes and temperatures and for very small pressures.

## 10.5 VIRIAL EXPANSIONS

The relations between  $p\bar{v}$  and  $p$  in the form of power series, as given in Eq. (10.1), may be expressed as

$$p\bar{v} = A(1 + B'p + C'p^2 + D'p^3 + \dots)$$

For any gas, from Eq. (10.3)

$$\lim_{p \rightarrow 0} p\bar{v} = A = \bar{R}T$$

$$\begin{aligned} \therefore \frac{p\bar{v}}{\bar{R}T} &= 1 + B'p + C'p^2 + D'p^3 + \dots \\ &= 1 + \frac{B'\bar{R}T}{p} + C' \left( \frac{\bar{R}T}{p} \right)^2 + D' \left( \frac{\bar{R}T}{p} \right)^3 + \dots \end{aligned} \quad (10.66)$$

An alternative expression is

$$\frac{p\bar{v}}{\bar{R}T} = 1 + \frac{B}{\bar{v}} + \frac{C}{\bar{v}^2} + \frac{D}{\bar{v}^3} + \dots \quad (10.67)$$

Both expressions in Eqs. (10.66) and (10.67) are known as *virial expansions* or virial equations of state, first introduced by the Dutch physicist, Kammerlingh Onnes,  $B'$ ,  $C'$ ,  $B$ ,  $C$ , etc. are called *virial coefficients*.  $B'$  and  $B$  are called second virial coefficients,  $C'$  and  $C$  are called third virial coefficients and so on. For a given gas, these coefficients are the functions of temperature only.

The ratio  $p\bar{v}/\bar{R}T$  is called the *compressibility factor*,  $Z$ . For an ideal gas  $Z = 1$ . The magnitude of  $Z$  for a certain gas at a particular pressure and temperature gives an indication of the extent of deviation of the gas from the ideal gas behaviour. The virial expansions become

$$Z = 1 + B'p + C'p^2 + D'p^3 + \dots \quad (10.68)$$

$$\text{and} \quad Z = 1 + \frac{B}{\bar{v}} + \frac{C}{\bar{v}^2} + \frac{D}{\bar{v}^3} + \dots \quad (10.69)$$

The relations between  $B'$ ,  $C'$  and  $B$ ,  $C$ , ... can be derived as given below.

$$\begin{aligned} \frac{p\bar{v}}{\bar{R}T} &= 1 + B'p + C'p^2 + D'p^3 + \dots \\ &= 1 + B' \left[ \frac{\bar{R}T}{\bar{v}} \left( 1 + \frac{B}{\bar{v}} + \frac{C}{\bar{v}^2} + \dots \right) \right] \\ &\quad + C' \left[ \left( \frac{\bar{R}T}{\bar{v}} \right)^2 \left( 1 + \frac{B}{\bar{v}} + \frac{C}{\bar{v}^2} + \dots \right)^2 \right] + \dots \\ &= 1 + \frac{B'\bar{R}T}{\bar{v}} + \frac{B'B\bar{R}T + C'(\bar{R}T)^2}{\bar{v}^2} \\ &= 1 + \frac{B'\bar{R}TC + C'(\bar{R}T)^2 + D'(\bar{R}T)^3}{\bar{v}^3} + \dots \end{aligned} \quad (10.70)$$

### LO 10.9

Derive virial equations of state in terms of compressibility factor  $Z$



Comparing this equation with Eq. (10.81) and rearranging

$$B' = \frac{B}{\bar{R}T}, C' = \frac{C - B^2}{(\bar{R}T)^2},$$

$$D' = \frac{D - 3BC + 2B^3}{(\bar{R}T)^3}, \text{ and so on}$$

Therefore,

$$\begin{aligned} Z &= \frac{p\bar{v}}{\bar{R}T} = 1 + B'p + C'p^2 + \dots \\ &= 1 + \frac{B}{\bar{R}T}p + \frac{C - B^2}{(\bar{R}T)^2}p^2 + \dots \end{aligned} \quad (10.71)$$

The terms  $B/\bar{v}$ ,  $C/\bar{v}^2$ , etc. of the virial expansion arise on account of molecular interactions. If no such interactions exist (at very low pressures)  $B = 0$ ,  $C = 0$ , etc. and  $Z = 1$  and  $p\bar{v} = \bar{R}T$ .

## 10.6 LAW OF CORRESPONDING STATES

For a certain gas, the compressibility factor  $Z$  is a function of  $p$  and  $T$  [Eq. (10.71)], and so a plot can be made of lines of constant temperature on coordinates of  $p$  and  $Z$  (Fig. 10.4). From this plot,  $Z$  can be obtained for any value of  $p$  and  $T$ , and the volume can then be obtained from the equation  $p\bar{v} = ZRT$ . The advantage of using  $Z$  instead of a direct plot of  $\bar{v}$  is a smaller range of values in plotting.

For each substance, there is a compressibility factor chart. It would be very convenient if one chart could be used for all substances. The general shapes of the vapour dome and of the constant temperature lines on the  $p$ - $\bar{v}$  plane are similar for all substances, although the scales may be different. This physical similarity can be exploited by using dimensionless properties called *reduced properties*. The reduced pressure is the ratio of the existing pressure to the critical pressure of the substance, and similarly for reduced temperature and reduced volume. Then

$$p_r = \frac{p}{p_c}, T_r = \frac{T}{T_c}, v_r = \frac{v}{v_c}$$

where subscript  $r$  denotes the reduced property, and subscript  $c$  denotes the property at the critical state.

At the same pressure and temperature the specific or molal volumes of different gases are different. However, it is found from experimental data that at the same reduced pressure and reduced temperature, the reduced volumes of different gases are approximately the same. Therefore, for all substances

### LO 10.10

Discuss the law of corresponding states and discuss the making of generalized compressibility chart

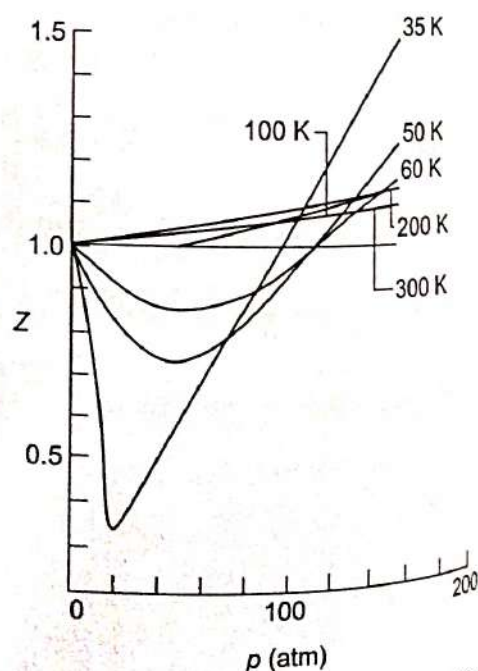


Fig. 10.4

Variation of the compressibility factor of hydrogen with pressure at constant temperature



$$v_r = f(p_r, T_r) \quad (10.72)$$

Now,

$$v_r = \frac{v}{v_c} = \frac{ZRTp_c}{Z_cRT_cp} = \frac{Z}{Z_c} \cdot \frac{T_r}{p_r} \quad (10.73)$$

where  $Z_c = \frac{p_c v_c}{RT_c}$ . This is called the critical compressibility factor. Therefore, from Eqs. (10.72) and (10.73),

$$Z = f(p_r, T_r, Z_c) \quad (10.74)$$

Experimental values of  $Z_c$  for most substances fall within a narrow range 0.20–0.30. Therefore,  $Z_c$  may be taken to be a constant and Eq. (10.74) reduces to

$$Z = f(p_r, T_r) \quad (10.75)$$

When  $T_r$  is plotted as a function of reduced pressure and  $Z$ , a single plot, known as the *generalised compressibility chart*, is found to be satisfactory for a great variety of substances. Although necessarily approximate, the plots are extremely useful in situations where detailed data on a particular gas are lacking but its critical properties are available.

The relation among the reduced properties,  $p_r$ ,  $T_r$  and  $v_r$ , is known as the *law of corresponding states*. It can be derived from the various equations of state, such as those of van der Waals, Berthelot and Dieterici. For a van der Waals gas,

$$\left(p + \frac{a}{v^2}\right)(v - b) = RT$$

where  $a$ ,  $b$  and  $R$  are the characteristic constants of the particular gas.

$$\therefore p = \frac{RT}{v - b} - \frac{a}{v^2}$$

$$\text{or } pv^3 - (pb + RT)v^2 + av - ab = 0$$

It is therefore a cubic in  $v$  and for given values of  $p$  and  $T$  has three roots of which only one need to be real. For low temperatures, three positive real roots exist for a certain range of pressure. As the temperature increases the three real roots approach one another and at the critical temperature they become equal. Above this temperature, only one real root exists for all values of  $p$ . The critical isotherm  $T_c$  at the critical state on the  $p$ - $v$  plane (Fig. 10.5), where the three real roots of the van der Waals equation coincide, not only has a zero slope, but also its slope changes at the critical state (point of inflection), so that the first and second derivatives of  $p$  with respect to  $v$  at  $T = T_c$  are each equal to zero. Therefore,

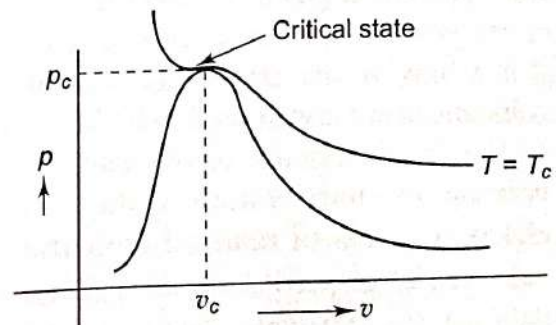


Fig. 10.5 Critical properties on  $p$ - $v$  diagram

$$\left(\frac{\partial p}{\partial v}\right)_{T=T_c} = -\frac{RT_c}{(v_c - b)^2} + \frac{2a}{v_c^3} = 0 \quad (10.76)$$



$$v_r = f(p_r, T_r)$$

Now,

$$v_r = \frac{v}{v_c} = \frac{ZRTp_c}{Z_cRT_c p} = \frac{Z}{Z_c} \cdot \frac{T_r}{p_r} \quad (10.72)$$

where  $Z_c = \frac{p_c v_c}{RT_c}$ . This is called the critical compressibility factor. Therefore, from Eqs. (10.72) and (10.73),

$$Z = f(p_r, T_r, Z_c) \quad (10.74)$$

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$$\left(p + \frac{a}{v^2}\right)(v - b) = RT$$

where  $a$ ,  $b$  and  $R$  are the characteristic constants of the particular gas.

$$\therefore p = \frac{RT}{v - b} - \frac{a}{v^2}$$

or

$$pv^3 - (pb + RT)v^2 + av - ab = 0$$

It is therefore a cubic in  $v$  and for given values of  $p$  and  $T$  has three roots of which only one need to be real. For low temperatures, three positive real roots exist for a certain range of pressure. As the temperature increases the three real roots approach one another and at the critical temperature they become equal. Above this temperature, only one real root exists for all values of  $p$ . The critical isotherm  $T_c$  at the critical state on the  $p-v$  plane (Fig. 10.5), where the three real roots of the van der Waals equation coincide, not only has a zero slope, but also its slope changes

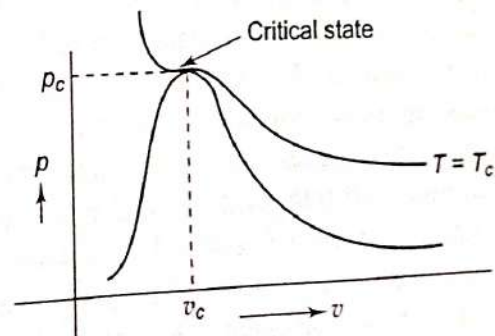


Fig. 10.5

Critical properties on  $p-v$  diagram

at the critical state (point of inflection), so that the first and second derivatives of  $p$  with respect to  $v$  at  $T = T_c$  are each equal to zero. Therefore,

$$\left(\frac{\partial p}{\partial v}\right)_{T=T_c} = -\frac{RT_c}{(v_c - b)^2} + \frac{2a}{v_c^3} = 0 \quad (10.76)$$



$$\left( \frac{\partial^2 p}{\partial v^2} \right)_{T=T_c} = \frac{2 \cdot RT_c}{(v_c - b)^3} - \frac{6a}{v_c^4} = 0 \quad (10.77)$$

From these two equations, by rearranging and dividing,  $b = \frac{1}{3} v_c$ .

Substituting the value of  $b$  in Eq. (10.76)

$$R = \frac{8a}{9T_c v_c}$$

Substituting the values of  $b$  and  $R$  in Eq. (10.60)

$$\left( p_c + \frac{a}{v_c^2} \right) \left( \frac{2}{3} v_c \right) = \frac{8a}{9T_c v_c} \cdot T_c$$

$$a = 3p_c v_c^2$$

Therefore, the value of  $R$  becomes

$$R = \frac{8}{3} \frac{p_c v_c}{T_c}$$

The values of  $a$ ,  $b$  and  $R$  have thus been expressed in terms of critical properties. Substituting these in the van der Waals equation of state

$$\left( p + \frac{3p_c v_c^2}{v^2} \right) \left( v - \frac{1}{3} v_c \right) = \frac{8}{3} \frac{p_c v_c}{T_c} \cdot T$$

or,

$$\left( \frac{p}{p_c} + \frac{3v_c^2}{v^2} \right) \left( \frac{v}{v_c} - \frac{1}{3} \right) = \frac{8}{3} \frac{T}{T_c}$$

Using the reduced parameters,

$$\therefore \left( p_r + \frac{3}{v_r^2} \right) (3v_r - 1) = 8T_r \quad (10.78)$$

In the reduced equation of state (10.78), the individual coefficients  $a$ ,  $b$  and  $R$  for a particular gas have disappeared. So this equation is an expression of the *law of corresponding states* because it reduces the properties of all gases to one formula. It is a 'law' to the extent that real gases obey van der Waals equation. Two different substances are considered to be in 'corresponding states', if their pressures, volumes and temperatures are of the same fractions (or multiples) of the critical pressure, volume and temperature of the two substances. The *generalised compressibility charts* in terms of reduced properties are shown in Fig. 10.6(a), (b) and (c). It is very useful in predicting the properties of substances for which more precise data are not available. The value of  $Z$  at the critical state of a van der Waals gas is 0.375 (since  $R = \frac{8}{3} \frac{p_c v_c}{T_c}$ ). At very low pressures,  $Z$  approaches unity as a real gas approaches the ideal gas behaviour. Equation (10.78) can also be written in the following form:

$$\left( p_r v_r + \frac{3}{v_r} \right) (3v_r - 1) = 8T_r v_r$$

$\therefore$

$$p_r v_r = \frac{8T_r v_r}{3v_r - 1} - \frac{3}{v_r} \quad (10.79)$$

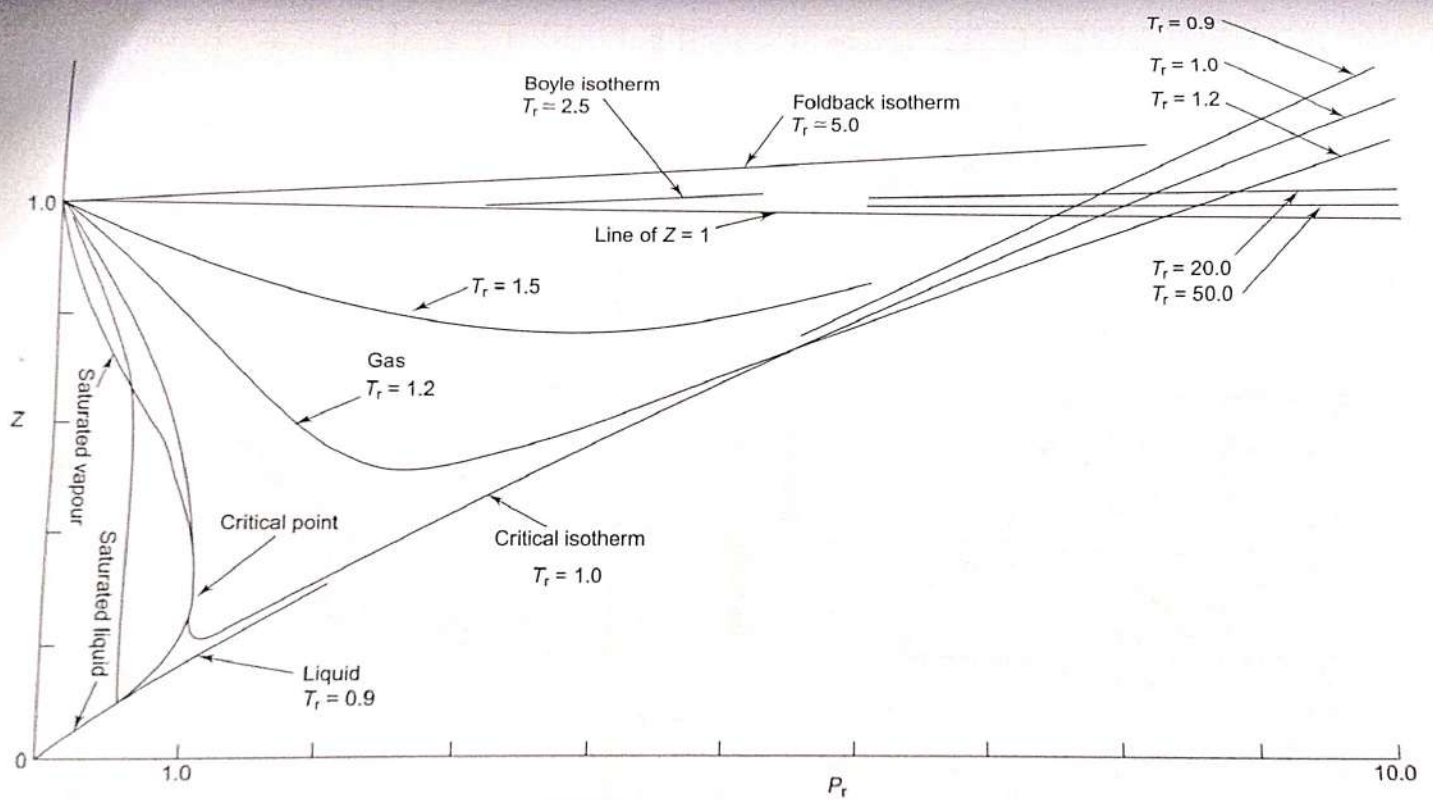
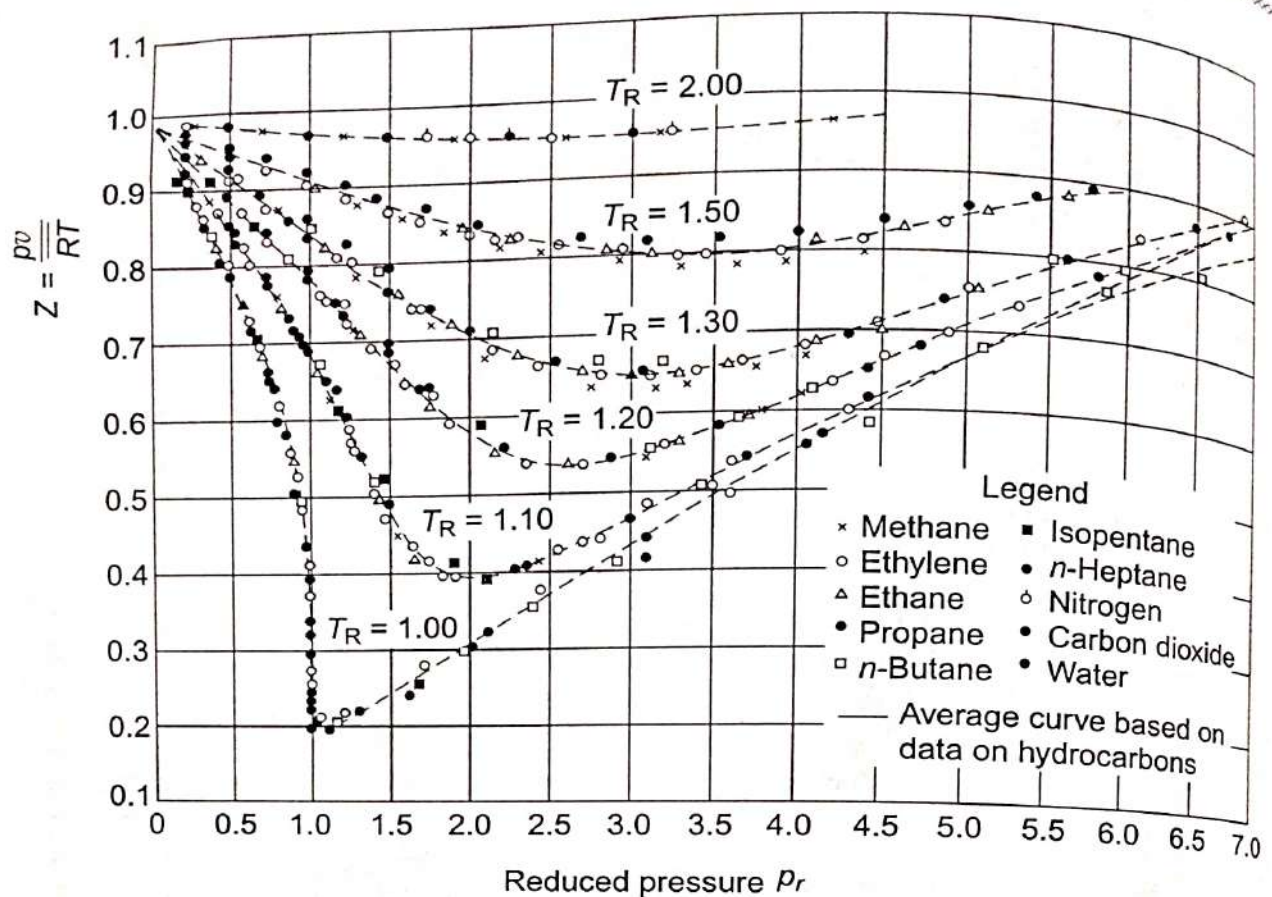


Fig. 10.6(a) Generalised compressibility chart





**Fig. 10.6(b)** Generalised compressibility chart

Figure 10.6 shows the law of corresponding states in reduced coordinates,  $(p_r, v_r)$  vs  $p_r$ . Differentiating Eq. (10.79) with respect to  $p_r$  and making it equal to zero, it is possible to determine the minima of the isotherms as shown below.

$$\frac{\partial}{\partial p_r} \left[ \frac{8T_r v_r}{3v_r - 1} - \frac{3}{v_r} \right]_{T_r} = 0$$

or

$$\frac{\partial}{\partial v_r} \left[ \frac{8T_r v_r}{3v_r - 1} - \frac{3}{v_r} \right]_{T_r} \left[ \frac{\partial v_r}{\partial p_r} \right]_{T_r} = 0$$

$$\frac{\partial}{\partial v_r} \left[ \frac{8T_r v_r}{3v_r - 1} - \frac{3}{v_r} \right]_{T_r} = 0$$

$\therefore$

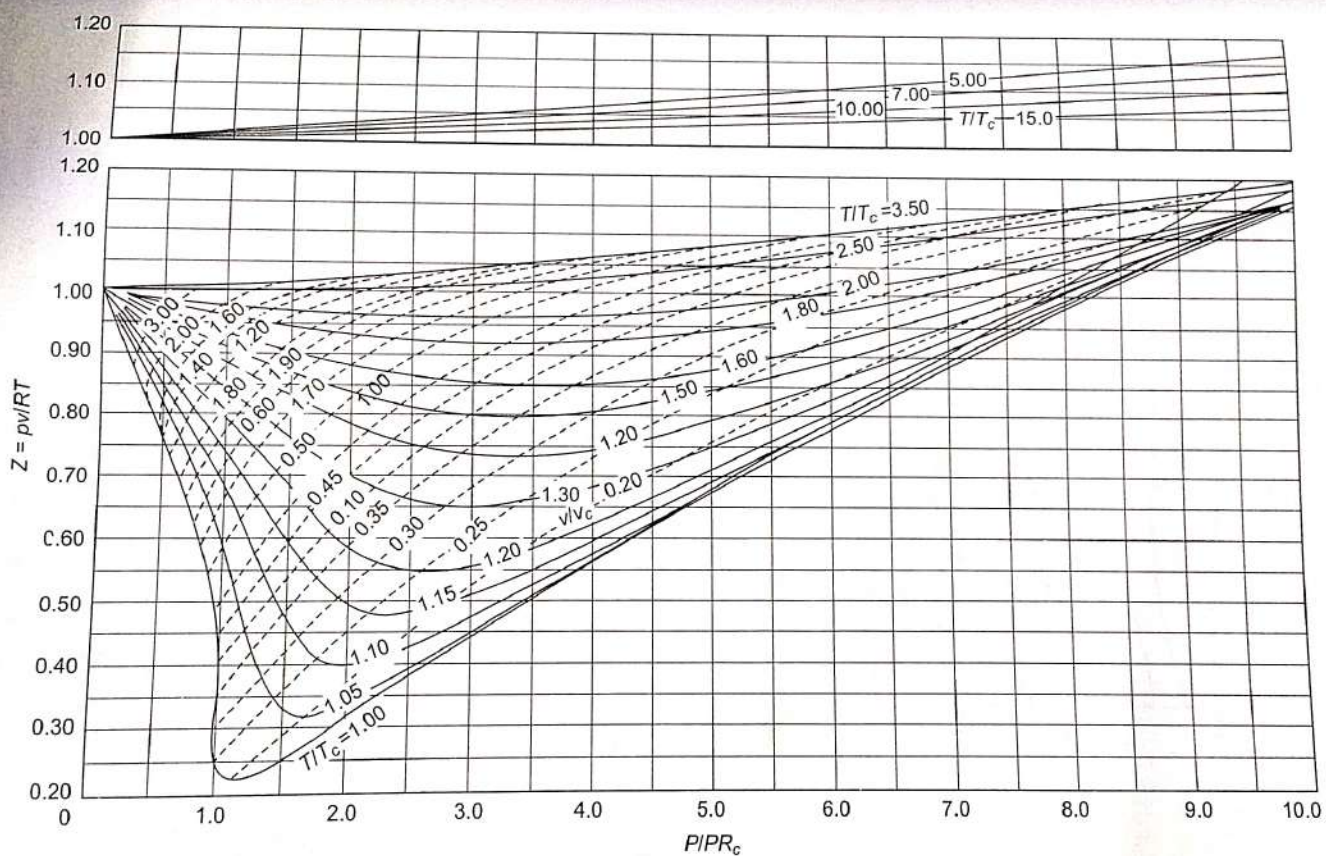
$$\frac{8T_r}{(3v_r - 1)^2} = \frac{3}{v_r^2}$$

or

$$\frac{3(3v_r - 1)^2}{v_r^2} = 8T_r = \left[ p_r + \frac{3}{v_r^2} \right] (3v_r - 1)$$

Simplifying

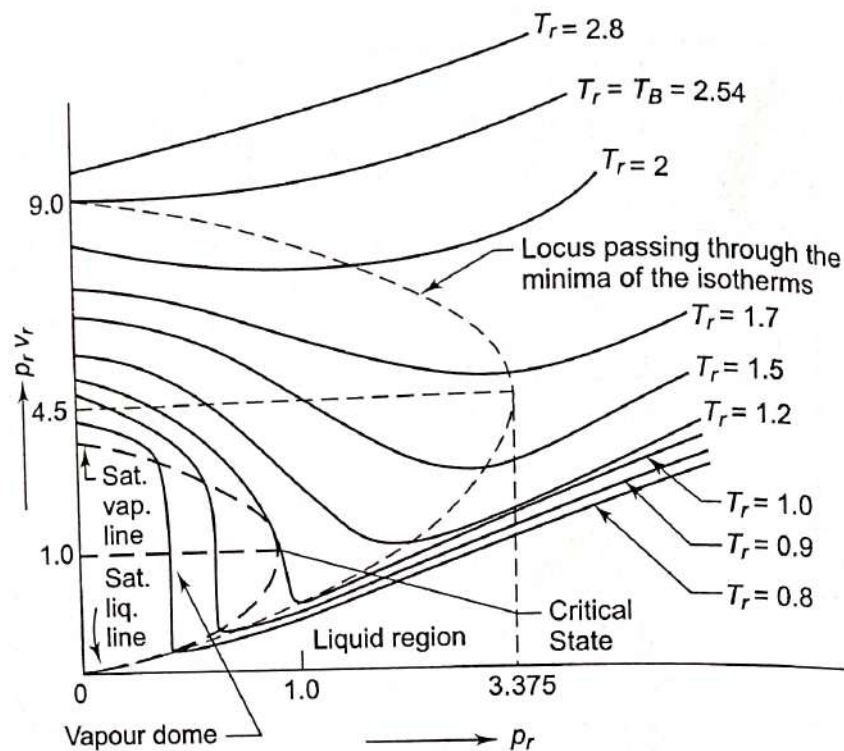
$$(p_r v_r)^2 - 9(p_r v_r) + 6p_r = 0$$



**Fig. 10.6(C)** Generalised compressibility chart—low pressure range. Data from L.C. Nelson and E.F. Obert, *Generalised compressibility charts*, Chem. En., vol. 61, p. 203, 1954. Note:  $v/v_c = p_c v/RT_c$



This is the equation of a parabola passing through the minima of the isotherms (Fig. 10.7).



**Fig. 10.7** Law of corresponding states in reduced coordinates

When

$$p_r = 0,$$

$$p_r v_r = 0, 9$$

Again

$$p_r = \frac{9(p_r v_r) - (p_r v_r)^2}{6}$$

$$\frac{dp_r}{d(p_r v_r)} = 9 - 2(p_r v_r) = 0$$

$\therefore$

$$p_r v_r = 4.5$$

$$p_r = \frac{9 \times 4.5 - (4.5)^2}{6} = 3.375$$

The parabola has the vertex at  $p_r v_r = 4.5$  and  $p_r = 3.375$ , and it intersects the ordinate at 0 and 9.

Each isotherm up to that marked  $T_B$  has a minimum (Fig. 10.7). The  $T_B$  isotherm has an initial horizontal portion ( $p_r v_r = \text{constant}$ ) so that Boyle's law is obeyed fairly accurately up to moderate pressures. Hence, the corresponding temperature is called the *Boyle temperature* for that gas. The Boyle temperature  $T_B$  can be determined by making

$$\left[ \frac{\partial(p_r v_r)}{\partial p_r} \right]_{T_r=T_B} = 0 \text{ when } p_r = 0$$

Above the Boyle temperature, the isotherms slope upwards and show no minima.

As  $T_r$  is reduced below the critical (i.e. for  $T_r < 1$ ), the gas becomes liquefied, and during phase transition isotherms are vertical. The minima of all these isotherms lie in the liquid zone.

van der Waals equation of state can be expressed in the virial form given as follows.

$$\left(p + \frac{a}{v^2}\right)(v - b) = RT$$

$$\left(pv + \frac{a}{v}\right)\left(1 - \frac{b}{v}\right) = RT$$

$$pv + \frac{a}{v} = RT\left(1 - \frac{b}{v}\right)^{-1}$$

$$= RT\left[1 + \frac{b}{v} + \frac{b^2}{v^2} + \frac{b^3}{v^3} + \dots\right] \quad \left(\text{where } \frac{b}{v} < 1\right)$$

$$pv = RT\left[1 + \left(b - \frac{a}{RT}\right)\frac{1}{v} + \frac{b^2}{v^2} + \frac{b^3}{v^3} + \dots\right] \quad (10.80)$$

$\therefore$  The second virial coefficient  $B = b - a/RT$ , the third virial coefficient  $C = b^2$ , etc. From Eq. (10.71), on mass basis

$$v = RT\left(1 + \frac{B}{RT}p + \frac{C - B^2}{RT^2}p^2 + \dots\right)$$

To determine Boyle temperature,  $T_B$

$$\left[\frac{\partial(pv)}{\partial p}\right]_{T=C, p=0} = 0 = \frac{B}{RT}$$

$$B = 0$$

or  $T_B = \frac{a}{bR}$ , because  $B = b - \frac{a}{RT}$

The point at which  $B$  is equal to zero gives the Boyle temperature. The second virial

coefficient is the most important. Since  $\left[\frac{\partial(pv)}{\partial p}\right]_{p=0} = B$ , when  $B$  is known, the

behaviour of the gas at moderate pressures is completely determined. The terms which contain higher powers ( $C/v^2$ ,  $D/v^3$ , etc.) becomes significant only at very high pressures.

## 10.7 OTHER EQUATIONS OF STATE

van der Waals equation is one of the oldest equations of state introduced in 1899, where the constants  $a$  and  $b$  are related to the critical state properties as found earlier,

$$a = 3p_c v_c^2 = \frac{27}{64} \frac{R^2 T_c^2}{p_c}, \quad b = \frac{1}{3} v_c = \frac{1}{8} \frac{RT_c}{p_c}$$

### LO 10.11

Explain other equations of state including van der Waals and Beattie-Bridgeman



The Beattie–Bridgeman equation developed in 1928 is given in Eq. (10.65) which has five constants. It is essentially an empirical curve fit of data, and is reasonably accurate when values of specific volume are greater than  $v_c$ .

Benedict, Webb and Rubin extended the Beattie–Bridgeman equation of state to cover a broader range of states as given below.

$$p = \frac{\bar{R}T}{v} + \left( B\bar{R}T - A - \frac{c}{T^2} \right) \frac{1}{\bar{v}^2} + \frac{(b\bar{R}T - a)}{\bar{v}^3} + \frac{a}{\bar{v}^6} + \frac{c}{\bar{v}^3 T^2} \left( 1 + \frac{\gamma}{\bar{v}^2} \right) \exp \left( -\frac{\gamma}{\bar{v}^2} \right)$$

It has eight constants and is quite successful in predicting the  $p$ – $v$ – $T$  behaviour of light hydrocarbons.

The Redlich–Kwong equation proposed in 1949 and given by Eq. (10.63) has the constants  $a$  and  $b$  in terms of critical properties as follows:

$$a = 0.4275 \frac{R^2 T_c^{2.5}}{p_c}, \quad b = 0.0867 \frac{RT_c}{p_c}$$

The values of the constants for the van der Waals, Redlich–Kwong and Benedict–Webb–Rubin equations of state are given in Table 10.2, while those for the Beattie–Bridgeman equation of state are given in Table 10.3. Apart from these, many other multi-constant equations of state have been proposed. With the advent of high speed computers, equations having 50 or more constants have been developed for representing the  $p$ – $v$ – $T$  behaviour of different substances.

**Table 10.2** Constants for the van der Waals, Redlich–Kwong and Benedict–Webb–Rubin equations of state

1. van der Waals and Redlich–Kwong Constants for pressure in bars, specific volume in  $\text{m}^3/\text{kmol}$  and temperature in K.

| Substance                            | van der Waals                                                       |                                         | Redlich–Kwong                                                               |                                         |
|--------------------------------------|---------------------------------------------------------------------|-----------------------------------------|-----------------------------------------------------------------------------|-----------------------------------------|
|                                      | $a$<br>$\text{bar} \left( \frac{\text{m}^3}{\text{kmol}} \right)^2$ | $b$<br>$\frac{\text{m}^3}{\text{kmol}}$ | $a$<br>$\text{bar} \left( \frac{\text{m}^3}{\text{kmol}} \right)^2 K^{1/2}$ | $b$<br>$\frac{\text{m}^3}{\text{kmol}}$ |
| Air                                  | 1.368                                                               | 0.0367                                  | 15.989                                                                      | 0.02541                                 |
| Butane ( $\text{C}_4\text{H}_{10}$ ) | 13.86                                                               | 0.1162                                  | 289.55                                                                      | 0.08060                                 |
| Carbon dioxide ( $\text{CO}_2$ )     | 3.647                                                               | 0.0428                                  | 64.43                                                                       | 0.02963                                 |
| Carbon monoxide (CO)                 | 1.474                                                               | 0.0395                                  | 17.22                                                                       | 0.02737                                 |
| Methane ( $\text{CH}_4$ )            | 2.293                                                               | 0.0428                                  | 32.11                                                                       | 0.02965                                 |
| Nitrogen ( $\text{N}_2$ )            | 1.366                                                               | 0.0386                                  | 15.53                                                                       | 0.02677                                 |
| Oxygen ( $\text{O}_2$ )              | 1.369                                                               | 0.317                                   | 17.22                                                                       | 0.02197                                 |
| Propane ( $\text{C}_3\text{H}_8$ )   | 9.349                                                               | 0.0901                                  | 182.23                                                                      | 0.06242                                 |
| Refrigerant 12                       | 10.49                                                               | 0.0971                                  | 208.59                                                                      | 0.06731                                 |
| Sulfur dioxide ( $\text{SO}_2$ )     | 6.883                                                               | 0.0569                                  | 144.80                                                                      | 0.03945                                 |
| Water ( $\text{H}_2\text{O}$ )       | 5.531                                                               | 0.0305                                  | 142.59                                                                      | 0.02111                                 |

Source: Calculated from critical data.



2. Benedict-Webb-Rubin Constants for pressure in bars, specific volume in  $\text{m}^3/\text{kmol}$  and temperature in K.

| Substance                 | $a$    | $A$    | $b$      | $B$     | $c$                 | $C$                 | $\alpha$               | $\gamma$ |
|---------------------------|--------|--------|----------|---------|---------------------|---------------------|------------------------|----------|
| $\text{C}_4\text{H}_{10}$ | 1.9073 | 10.218 | 0.039998 | 0.12436 | $3.206 \times 10^5$ | $1.006 \times 10^6$ | $1.101 \times 10^{-3}$ | 0.0340   |
| $\text{CO}_2$             | 0.1386 | 2.7737 | 0.007210 | 0.04991 | $1.512 \times 10^4$ | $1.404 \times 10^5$ | $8.47 \times 10^{-5}$  | 0.00539  |
| $\text{CO}$               | 0.0371 | 1.3590 | 0.002632 | 0.05454 | $1.054 \times 10^3$ | $8.676 \times 10^3$ | $1.350 \times 10^{-4}$ | 0.0060   |
| $\text{CH}_4$             | 0.0501 | 1.8796 | 0.003380 | 0.04260 | $2.579 \times 10^3$ | $2.287 \times 10^4$ | $1.244 \times 10^{-4}$ | 0.0060   |
| $\text{N}_2$              | 0.0254 | 1.0676 | 0.002328 | 0.04074 | $7.381 \times 10^2$ | $8.166 \times 10^1$ | $1.272 \times 10^{-4}$ | 0.0053   |

Source: H.W. Copper, and J.C. Goldfrank, Hydrocarbon Processing, 45 (12); 141 (1967).

Table 10.3

- (a) The Beattie-Bridgeman equation of state is

$$p = \frac{R_u T}{\bar{v}^2} \left( 1 - \frac{c}{\bar{v} T^2} \right) (\bar{v} + B) - \frac{A}{\bar{v}}, \text{ where } A = A_0 \left( 1 - \frac{a}{\bar{v}} \right) \text{ and } B = B_0 \left( 1 - \frac{b}{\bar{v}} \right)$$

When  $p$  is in kPa,  $\bar{v}$  is in  $\text{m}^3/\text{kmol}$ ,  $T$  is in K and  $R_u = 8.314 \text{ kPa} \cdot \text{m}^3/(\text{kmol} \cdot \text{K})$ , the five constants in the Beattie-Bridgeman equation are as follows:

| Gas                           | $A_0$    | $a$      | $B_0$   | $b$       | $c$                |
|-------------------------------|----------|----------|---------|-----------|--------------------|
| Air                           | 131.8441 | 0.01931  | 0.04611 | -0.001101 | $4.34 \times 10^4$ |
| Argon, Ar                     | 130.7802 | 0.02328  | 0.03931 | 0.0       | $5.99 \times 10^4$ |
| Carbon dioxide, $\text{CO}_2$ | 507.2836 | 0.07132  | 0.10476 | 0.07235   | $6.60 \times 10^5$ |
| Helium, He                    | 2.1886   | 0.05984  | 0.01400 | 0.0       | 40                 |
| Hydrogen, $\text{H}_2$        | 20.0117  | -0.00506 | 0.02096 | -0.04359  | 504                |
| Nitrogen, $\text{N}_2$        | 136.2315 | 0.02617  | 0.05046 | -0.00691  | $4.20 \times 10^4$ |
| Oxygen, $\text{O}_2$          | 151.0857 | 0.02562  | 0.04624 | 0.004208  | $4.80 \times 10^4$ |

Source: Gordon J. Van Wylen and Richard E. Sonntag, Fundamentals of Classical Thermodynamics, English/SI Version, 3d Edition., Wiley New York, 1986, p. 46. Table 3.3.

## 10.8 PROPERTIES OF MIXTURES OF GASES— DALTON'S LAW OF PARTIAL PRESSURES

Let us imagine a homogeneous mixture of inert ideal gases at a temperature  $T$ , a pressure  $p$  and a volume  $V$ . Let us suppose there are  $n_1$  moles of gas  $A_1$ ,  $n_2$  moles of gas  $A_2$ , ... and up to  $n_c$  moles of gas  $A_c$  (Fig. 10.8). Since there is no chemical reaction, the mixture is in a state of equilibrium with the equation of state

$$pV = (n_1 + n_2 + \dots n_c) \bar{R}T$$

where

$$\bar{R} = 8.3143 \text{ kJ/kg mol K}$$

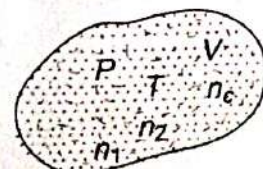


Fig. 10.8 Mixture of gases

### LO 10.12

Explain Dalton's law of partial pressure and Amagat's law of partial volume in a gas mixture



$$\therefore p = \frac{n_1 \bar{R}T}{V} + \frac{n_2 \bar{R}T}{V} + \dots + \frac{n_c \bar{R}T}{V}$$

The expression  $\frac{n_k \bar{R}T}{V}$  represents the pressure that the  $k$ th gas would exert if it occupied the volume  $V$  alone at temperature  $T$ . This is called the *partial pressure* of the  $k$ th gas and is denoted by  $p_k$ . Thus,

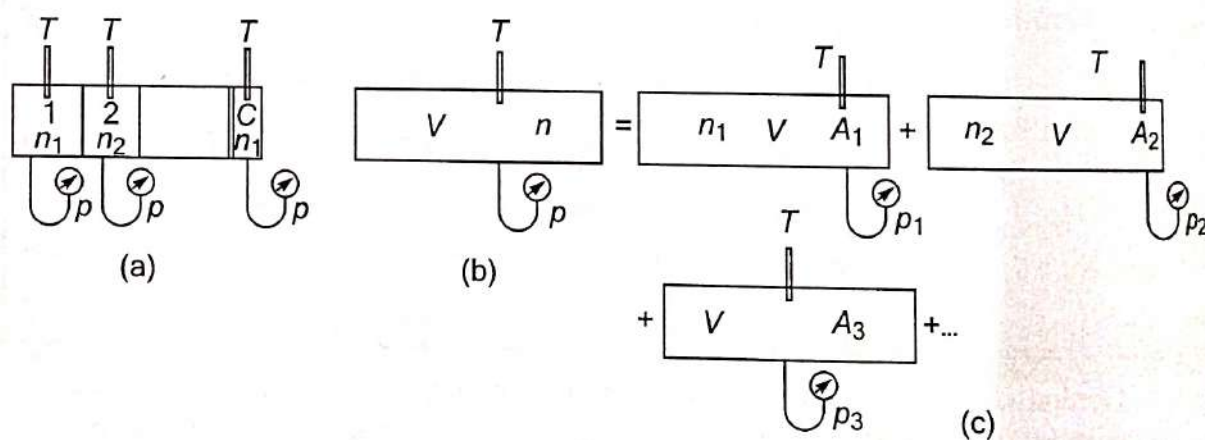
$$p_1 = \frac{n_1 \bar{R}T}{V}, p_2 = \frac{n_2 \bar{R}T}{V}, \dots, p_c = \frac{n_c \bar{R}T}{V}$$

and

$$p = p_1 + p_2 + \dots + p_c \quad (10.81)$$

This is known as *Dalton's law of partial pressures* which states that the total pressure of a mixture of ideal gases is equal to the sum of the partial pressures.

Visualisation of Dalton's law of partial pressures is shown in Fig. 10.8. There are  $n_1$  moles of gas  $A_1$  in compartment 1,  $n_2$  moles of gas  $A_2$  in compartment 2 and so on, each at pressure  $p$  and temperature  $T$ . Compartments are separated from one another by partitions shown in (a). When the partitions are removed, the gases mix and diffuse into one another till equilibrium is reached as shown in (b). If each of these gases  $A_1, A_2$ , etc. Occupies the entire volume of the mixture at the same temperature, the pressure gauges will record the respective partial pressures shown in (c) and Dalton's states that  $p = p_1 + p_2 + \dots + p_c$ .



**Fig. 10.8** Demonstration of two of partial pressures

Now

$$\begin{aligned} V &= (n_1 + n_2 + \dots + n_c) \cdot \frac{\bar{R}T}{p} \\ &= \sum n_K \cdot \frac{\bar{R}T}{p} \end{aligned}$$

and the partial pressure of the  $k$ th gas is

$$p_k = \frac{n_k \bar{R}T}{V}$$

Substituting the value of  $V$

$$p_K = \frac{n_K \cdot p}{\sum n_K} = \frac{n_K}{\sum n_K} \cdot p$$

Now

$$\Sigma n_K = n_1 + n_2 + \dots + n_c$$

= Total number of moles of gas

The ratio  $\frac{n_k}{\Sigma n_K}$  is called the *mole fraction of the kth gas*, and is denoted by  $x_K$ .

Thus

$$x_1 = \frac{n_1}{\Sigma n_K}, x_2 = \frac{n_2}{\Sigma n_K}, \dots, x_c = \frac{n_c}{\Sigma n_K}$$

and

$$p_1 = x_1 p, p_2 = x_2 p, \dots, p_c = x_c p$$

or

$$p_K = x_K \cdot p \quad (10.82)$$

Also

$$x_1 + x_2 + \dots + x_c = 1 \quad (10.83)$$

In a mixture of gases, if not all but one mole fraction is determined, the last can be calculated from the above equation. Again, in terms of masses

$$p_1 V = m_1 R_1 T$$

$$p_2 V = m_2 R_2 T$$

.....

$$p_c V = m_c R_c T$$

Adding, and using Dalton's law

$$pV = (m_1 R_1 + m_2 R_2 + \dots + m_c R_c) T \quad (10.84)$$

where

$$p = p_1 + p_2 + \dots + p_c$$

For the gas mixture

$$pV = (m_1 + m_2 + \dots + m_c) R_m T \quad (10.85)$$

where  $R_m$  is the gas constant for the mixture. From Eqs. (10.84) and (10.85)

$$R_m = \frac{m_1 R_1 + m_2 R_2 + \dots + m_c R_c}{m_1 + m_2 + \dots + m_c} \quad (10.86)$$

The gas constant of the mixture is thus the weighted mean, on a mass basis, of the gas constants of the components.

The total mass of gas mixture  $m$  is

$$m = m_1 + \dots + m_c$$

If  $\mu$  denotes the *equivalent molecular weight* of the mixture having  $n$  total number of moles.

$$n\mu = n_1 \mu_1 + n_2 \mu_2 + \dots + n_c \mu_c$$

$\therefore$

$$\mu = x_1 \mu_1 + x_2 \mu_2 + \dots + x_c \mu_c$$

or

$$\mu = \Sigma x_k \mu_K \quad (10.87)$$

The atmospheric air is assumed to consist of 21 percent  $O_2$ , 78 per cent  $N_2$  and 1 percent  $A$  by volume. Its molecular weight is

$$\begin{aligned} \mu_{\text{air}} &= 0.78 \times 28 + 0.21 \times 32 + 0.01 \times 40 \\ &= 21.84 + 6.72 + 0.4 = 28.96 \end{aligned}$$



A quantity called the *partial volume* of a component of a mixture is the volume that the component alone would occupy at the pressure and temperature of the mixture. Designating the partial volumes by  $V_1, V_2$ , etc.

$$pV_1 = m_1 R_1 T, pV_2 = m_2 R_2 T, \dots, pV_c = m_c R_c T$$

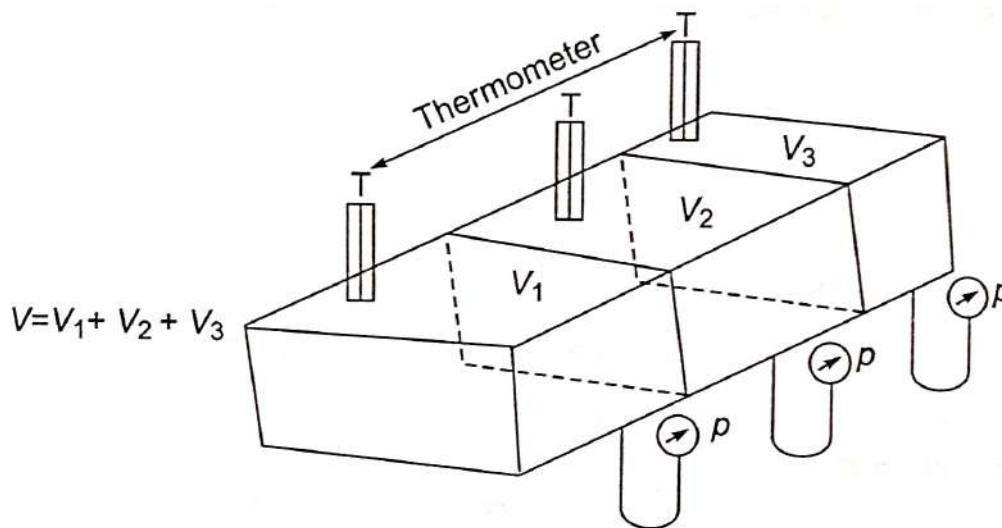
$$\text{or} \quad p(V_1 + V_2 + \dots + V_c) = (m_1 R_1 + m_2 R_2 + \dots + m_c R_c) T \quad (10.88)$$

From Eqs. (10.84), (10.85) and (10.88)

$$V = V_1 + V_2 + \dots + V_c \quad (10.89)$$

The total volume is thus equal to the sum of the partial volumes.

This is known as *Amagat's law of additive volumes*. Figure 10.9 shows that each component in a gas mixture occupies the respective partial volume at the same pressure  $p$  and temperature  $T$  of the mixture and Amagat's law states that the total volume of the mixture is equal to the sum of the partial volumes.



**Fig. 10.9** Visualisation of Amagat's law of additive volumes

The specific volume of the mixture,  $v$ , is given by

$$v = \frac{V}{m} = \frac{V}{m_1 + m_2 + \dots + m_c}$$

$$\begin{aligned} \text{or} \quad \frac{1}{v} &= \frac{m_1 + m_2 + \dots + m_c}{V} \\ &= \frac{m_1}{V} + \frac{m_2}{V} + \dots + \frac{m_c}{V} \end{aligned}$$

$$\text{or} \quad \frac{1}{v} = \frac{1}{v_1} + \frac{1}{v_2} + \dots + \frac{1}{v_c} \quad (10.90)$$

where  $v_1, v_2, \dots$  denote specific volumes of the components, each component occupying the total volume.

Therefore, the density of the mixture

$$\rho = \rho_1 + \rho_2 + \dots + \rho_c \quad (10.91)$$

## 10.9 INTERNAL ENERGY, ENTHALPY AND SPECIFIC HEATS OF GAS MIXTURES

When gases at equal pressures and temperatures are mixed adiabatically without work, as by inter-diffusion in a constant volume container, the first law requires that the internal energy of the gaseous system remains constant, and experiments show that the temperature remains constant. Hence, the internal energy of a mixture of gases is equal to the sum of the internal energies of the individual components, each taken at the temperature and volume of the mixture (i.e. sum of the 'partial' internal energies). This is also true for any of the thermodynamic properties like  $H$ ,  $C_v$ ,  $C_p$ ,  $S$ ,  $F$  and  $G$ , and is known as *Gibbs theorem*. Therefore, on a mass basis

$$mu_m = m_1u_1 + m_2u_2 + \dots + m_cu_c$$

$$u_m = \frac{m_1u_1 + m_2u_2 + \dots + m_cu_c}{m_1 + m_2 + \dots + m_c} \quad (10.92)$$

which is the average specific internal energy of the mixture.

Similarly, the total enthalpy of a gas mixture is the sum of the 'partial' enthalpies

$$mh_m = m_1h_1 + m_2h_2 + \dots + m_ch_c$$

$$h_m = \frac{m_1h_1 + m_2h_2 + \dots + m_ch_c}{m_1 + m_2 + \dots + m_c} \quad (10.93)$$

From the definitions of specific heats, it follows that

$$c_{vm} = \frac{m_1c_{v1} + m_2c_{v2} + \dots + m_cc_{vc}}{m_1 + m_2 + \dots + m_c} \quad (10.94)$$

and

$$c_{pm} = \frac{m_1c_{p1} + m_2c_{p2} + \dots + m_cc_{pc}}{m_1 + m_2 + \dots + m_c} \quad (10.95)$$

## 10.10 ENTROPY OF GAS MIXTURES

Gibbs theorem states that the total entropy of a mixture of gases is the sum of the partial entropies. The partial entropy of one of the gases of a mixture is the entropy that the gas would have if it occupied the whole volume alone at the same temperature.

Let us imagine a number of inert ideal gases separated from one another by suitable partitions, all the gases being at the same temperature  $T$  and pressure  $p$ . The total entropy (initial)

$$S_i = n_1s_1 + n_2s_2 + \dots + n_cs_c$$

$$= \sum n_K s_K$$

From property relation

$$Tds = dh - vdp = c_p dT - vdp$$

$$\therefore d\bar{s} = \bar{c}_p \frac{dT}{T} - \bar{R} \frac{dp}{p}$$



The entropy of 1 mole of the  $k$ th gas at  $T$  and  $p$

$$\bar{s}_K = \int c_{pK} \frac{dT}{T} - \bar{R} \ln p + \bar{s}_{0K}$$

where  $\bar{s}_{0K}$  is the constant of integration.

$$\therefore S_i = \bar{R} \sum n_K \left( \frac{1}{\bar{R}} \int \bar{c}_{pK} \frac{dT}{T} + \frac{\bar{s}_{0K}}{\bar{R}} - \ln p \right)$$

Let  
then

$$\sigma_K = \frac{1}{\bar{R}} \int \bar{c}_{pK} \frac{dT}{T} + \frac{\bar{s}_{0K}}{\bar{R}}$$

$$S_i = \bar{R} \sum n_K (\sigma_K - \ln p) \quad (10.96)$$

After the partitions are removed, the gases diffuse into one another at the same temperature and pressure, and by Gibbs theorem, the entropy of the mixture,  $S_f$ , is the sum of the partial entropies, with each gas exerting its respective partial pressure. Thus,

$$S_f = \bar{R} \sum n_K (\sigma_K - \ln p_K)$$

Since

$$p_K = x_K p$$

$$S_f = \bar{R} \sum n_K (\sigma_K - \ln x_K - \ln p) \quad (10.97)$$

A change in entropy due to the diffusion of any number of inert ideal gases is

$$S_f - S_i = -\bar{R} \sum n_K \ln x_K \quad (10.98)$$

or

$$S_f - S_i = -\bar{R} (n_1 \ln x_1 + n_2 \ln x_2 + \dots + n_c \ln x_c)$$

Since the mole fractions are less than unity,  $(S_f - S_i)$  is always positive, conforming to the second law.

Again

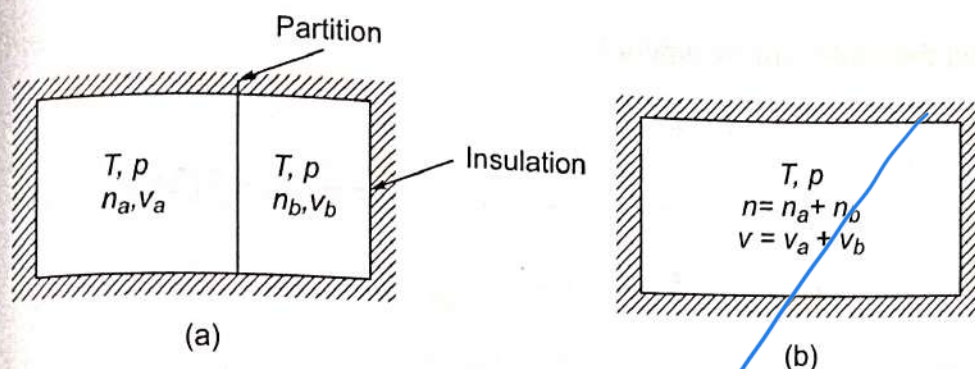
$$S_f - S_i = \left( n_1 \ln \frac{p_1}{p} + n_2 \ln \frac{p_2}{p} + \dots + n_c \ln \frac{p_c}{p} \right) \quad (10.99)$$

which indicates that each gas undergoes in the diffusion process a *free expansion* from total pressure  $p$  to the respective partial pressure at constant temperature.

Similarly, on a mass basis, the entropy change due to diffusion

$$\begin{aligned} S_f - S_i &= -\sum m_K R_K \ln \frac{p_K}{p} \\ &= -\left( m_1 R_1 \ln \frac{p_1}{p} + m_2 R_2 \ln \frac{p_2}{p} + \dots + m_c R_c \ln \frac{p_c}{p} \right) \end{aligned}$$

Let us consider that there are  $n_a$  moles of the first component of a gas mixture in the first compartment and  $n_b$  moles of the second component in the second component, both are at the same pressure and temperature (Fig. 10.10). The partition separating the components is removed and the gases mix adiabatically. We are to determine the entropy change due to mixing of the gases.



**Fig. 10.10** Entropy change due to diffusion of gases

For constant temperature, the change of entropy is

$$S_2 - S_1 = -n\bar{R} \ln \frac{p_2}{p_1}$$

and for the component 'a', the final pressure is  $p_a$ , its partial pressure, and the initial pressure is  $p$ . Therefore,

$$(S_2 - S_1)_a = -n_a \bar{R} \ln \frac{p_a}{p} = -n_a \bar{R} \ln x_a$$

Similarly, for the second component 'b'

$$(S_2 - S_1)_b = -n_b \bar{R} \ln \frac{p_b}{p} = -n_b \bar{R} \ln x_b$$

where  $x$  is the mole fraction.

So, the entropy change of the mixture is the sum of the entropy changes of the components

$$S_2 - S_1 = -n_a \bar{R} \ln x_a - n_b \bar{R} \ln x_b$$

If the components  $a$  and  $b$  are the same gas, then the entropy change in mixing is zero. This is because we cannot distinguish between the gases to determine the partial pressures, and the final pressure therefore is also the partial pressure of the gas. Thus, for *distinguishable* components, the entropy change due to mixing of any number of components is

$$S_2 - S_1 = -\bar{R} \sum_i n_i \ln x_i$$

which is always positive.

## 10.11 GIBBS FUNCTION OF A MIXTURE OF INERT IDEAL GASES

From the equations

$$d\bar{h} = \bar{c}_p dT$$

$$d\bar{s} = \bar{c}_p \frac{dT}{T} - \bar{R} \frac{dp}{p}$$

the enthalpy and entropy of 1 mole of an ideal gas at temperature  $T$  and pressure  $p$  are

$$\bar{h} = \bar{h}_0 + \int \bar{c}_p dT$$

$$\bar{s} = \int \bar{c}_p \frac{dT}{T} + \bar{s}_0 - \bar{R} \ln p$$



Therefore, the molar Gibbs function

$$\begin{aligned}\bar{g} &= \bar{h} - Ts \\ &= \bar{h}_0 + \int \bar{c}_p dT - T \int \bar{c}_p \frac{dT}{T} - T\bar{s}_0 + \bar{R}T \ln p\end{aligned}$$

Now  $\int d(uv) = \int u dv + \int v du = uv$

Let  $u = \frac{1}{T}, v = \int c_p dT$

Then 
$$\begin{aligned}\frac{1}{T} \int c_p dT &= \int \frac{1}{T} c_p dT + \int c_p dT \left( -\frac{1}{T^2} \right) dT \\ &= \int \frac{1}{T} c_p dT - \int \frac{c_p dT}{T^2}\end{aligned}$$

$$\int c_p dT - T \int c_p \frac{dT}{T} = -T \int \frac{c_p dT}{T^2}$$

Therefore,

$$\begin{aligned}\bar{g} &= \bar{h}_0 - T \int \frac{\bar{c}_p dT}{T^2} - T\bar{s}_0 + \bar{R}T \ln p \\ &= \bar{R}T \left( \frac{\bar{h}_0}{\bar{R}T} - \frac{1}{\bar{R}} \int \frac{\bar{c}_p dT}{T^2} - \frac{\bar{s}_0}{\bar{R}} + \ln p \right)\end{aligned}$$

Let

$$\phi = \frac{\bar{h}_0}{\bar{R}T} - \frac{1}{\bar{R}} \int \frac{\bar{c}_p dT}{T^2} - \frac{\bar{s}_0}{\bar{R}} \quad (10.100)$$

Thus,

$$\bar{g} = \bar{R}T(\phi + \ln p) \quad (10.101)$$

where  $\phi$  is a function of temperature only.

Let us consider a number of inert ideal gases separated from one another at the same  $T$  and  $p$

$$\begin{aligned}G_i &= \sum n_K g_K \\ &= \bar{R}T \sum n_K (\phi_K + \ln p)\end{aligned}$$

After the partitions are removed, the gases will diffuse, and the partial Gibbs function of a particular gas is the value of  $G$ , if that gas occupies the same volume at the same temperature exerting a partial pressure  $p_K$ . Thus,

$$\begin{aligned}G_f &= \bar{R}T \sum n_K (\phi_K + \ln p_K) \\ &= \bar{R}T \sum n_K (\phi_K + \ln p + \ln x_K)\end{aligned}$$

Therefore,

$$G_f - G_i = \bar{R}T \sum n_K \ln x_K \quad (10.102)$$

Since  $x_K < 1$ ,  $(G_f - G_i)$  is negative because  $G$  decreases due to diffusion. Gibbs function of a mixture of ideal gases at  $T$  and  $p$  is thus

$$G = \bar{R}T \sum n_K (\phi_K + \ln p + \ln x_K) \quad (10.103)$$



$$\therefore v = \frac{0.7 \times 8.3143 \times 57.85}{20.183 \times 5.46 \times 10^3} = 3.05 \times 10^{-3} \text{ m}^3/\text{kg}$$

$$\therefore v_r = \frac{v}{v_c} = \frac{3.05 \times 10^{-3} \times 20.183}{4.16 \times 10^{-2}} = 1.48$$

**Example 10.22** For the Berthelot equation of state

$$p = \frac{RT}{v-b} - \frac{a}{Tv^2}$$

show that (a)  $\lim_{\substack{p \rightarrow 0 \\ T \rightarrow \infty}} (RT - pv) = 0$

(b)  $\lim_{T \rightarrow \infty} \frac{v}{T} = \frac{R}{p}$ ,

(c) Boyle temperature,  $T_B = \sqrt{\frac{a}{bR}}$ ,

(d) Critical properties  $p_c = \frac{1}{12b} \sqrt{\frac{2aR}{3b}}$ ,  $v_c = 3b$ ,  $T_c = \sqrt{\frac{8a}{27bR}}$ ,

(e) Law of corresponding states

$$\left( p_r + \frac{3}{T_r \times v_r^2} \right) (3v_r - 1) = 8T_r$$

[LO 10.8, 10.9]

**Solution** (a)

$$p = \frac{RT}{v-b} - \frac{a}{Tv^2}$$

$$\therefore RT = \left( p + \frac{a}{Tv^2} \right) (v-b)$$

or  $\frac{RT}{p} = v + \frac{a}{pvT} - b - \frac{ab}{pv^2T}$

$$\therefore RT - pv = \frac{a}{vT} - bp - \frac{ab}{v^2T}$$

Proved (a)

$$\therefore \lim_{\substack{p \rightarrow 0 \\ T \rightarrow \infty}} (RT - pv) = 0$$

(b) Now  $v = \frac{RT}{p} - \frac{a}{pvT} + b + \frac{ab}{pv^2T}$

$$\therefore \frac{v}{T} = \frac{R}{p} - \frac{a}{pvT^2} + \frac{b}{T} + \frac{ab}{pv^2T^2}$$

Proved (b)

$$\therefore \lim_{T \rightarrow \infty} \frac{v}{T} = \frac{R}{p}$$

(c)  $pv = RT - \frac{a}{vT} + bp + \frac{ab}{v^2T}$

The last three terms of the equation are very small, except at very high pressures and small volume. Hence, substituting  $v = RT/p$

$$pv = RT - \frac{ap}{RT^2} + bp + \frac{abp^2}{R^2T^3}$$

$$\therefore \left[ \frac{\partial(pv)}{\partial p} \right]_T = -\frac{a}{RT^2} + b + \frac{2abp}{R^2T^3} = 0$$

When  $p = 0$ ,  $T = T_B$ , the Boyle temperature

$$\therefore \frac{a}{RT_B^2} = b$$

$$\text{or } T_B = \sqrt{\frac{a}{bR}}$$

Proved (c)

$$(d) \quad p = \frac{RT}{v-b} - \frac{a}{Tv^2}$$

$$\left( \frac{\partial p}{\partial v} \right)_{T=T_c} = -\frac{RT_c}{(v_c-b)^2} + \frac{2a}{T_c \cdot v_c^3} = 0$$

$$\left( \frac{\partial^2 p}{\partial v^2} \right)_{T=T_c} = \frac{2RT_c}{(v_c-b)^3} - \frac{6a}{T_c \cdot v_c^4} = 0$$

$$\left( p_c + \frac{a}{T_c v_c^2} \right) (v_c - b) = RT_c$$

By solving the three equations, as was done in the case of van der Waals equation of state in Section 10.7

$$p_c = \frac{1}{12b} \sqrt{\frac{2aR}{3b}}, v_c = 3b, \text{ and } T_c = \sqrt{\frac{8a}{27bR}}$$

Proved (d)

(e) Solving the above three equations

$$a = \frac{8v_c^3 p_c^2}{R} = 3p_c \cdot v_c^2 \cdot T_c$$

$$b = \frac{v_c}{3}, R = \frac{8}{3} \frac{p_c v_c}{T_c} \text{ (so that } Z_c = 3/8)$$

Substituting in the equation

$$\left( p + \frac{a}{Tv^2} \right) (v - b) = RT$$

$$\left( p + \frac{3p_c v_c^2 T_c}{Tv^2} \right) \left( v - \frac{v_c}{3} \right) = \frac{8p_c v_c}{3T_c} \cdot T$$

$$\left( p_r + \frac{3}{T_r v_r^2} \right) (3v_r - 1) = 8T_r$$

This is the law of corresponding states.

Proved (e)